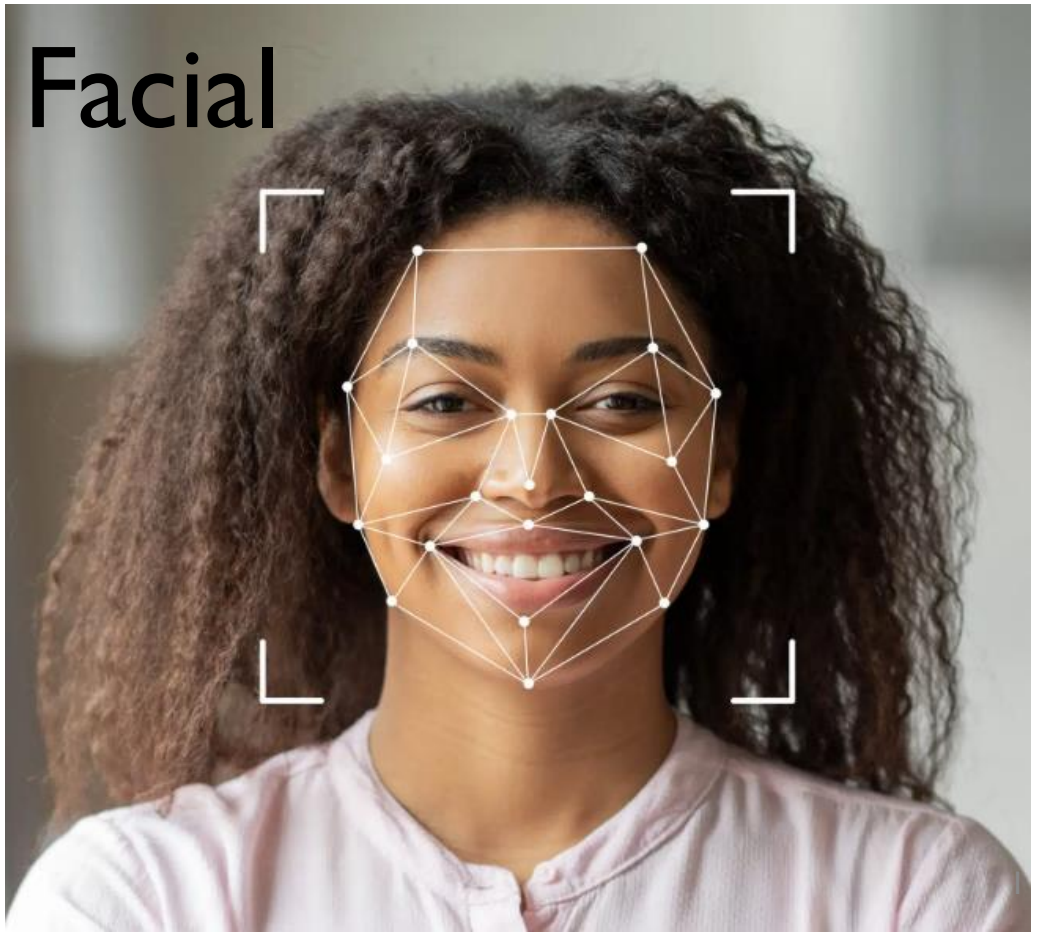
Visión por Computador

Versión 2026



Deep Learning: Análisis Facial

(CAPÍTULO 3 – CV03D)



Domingo Mery

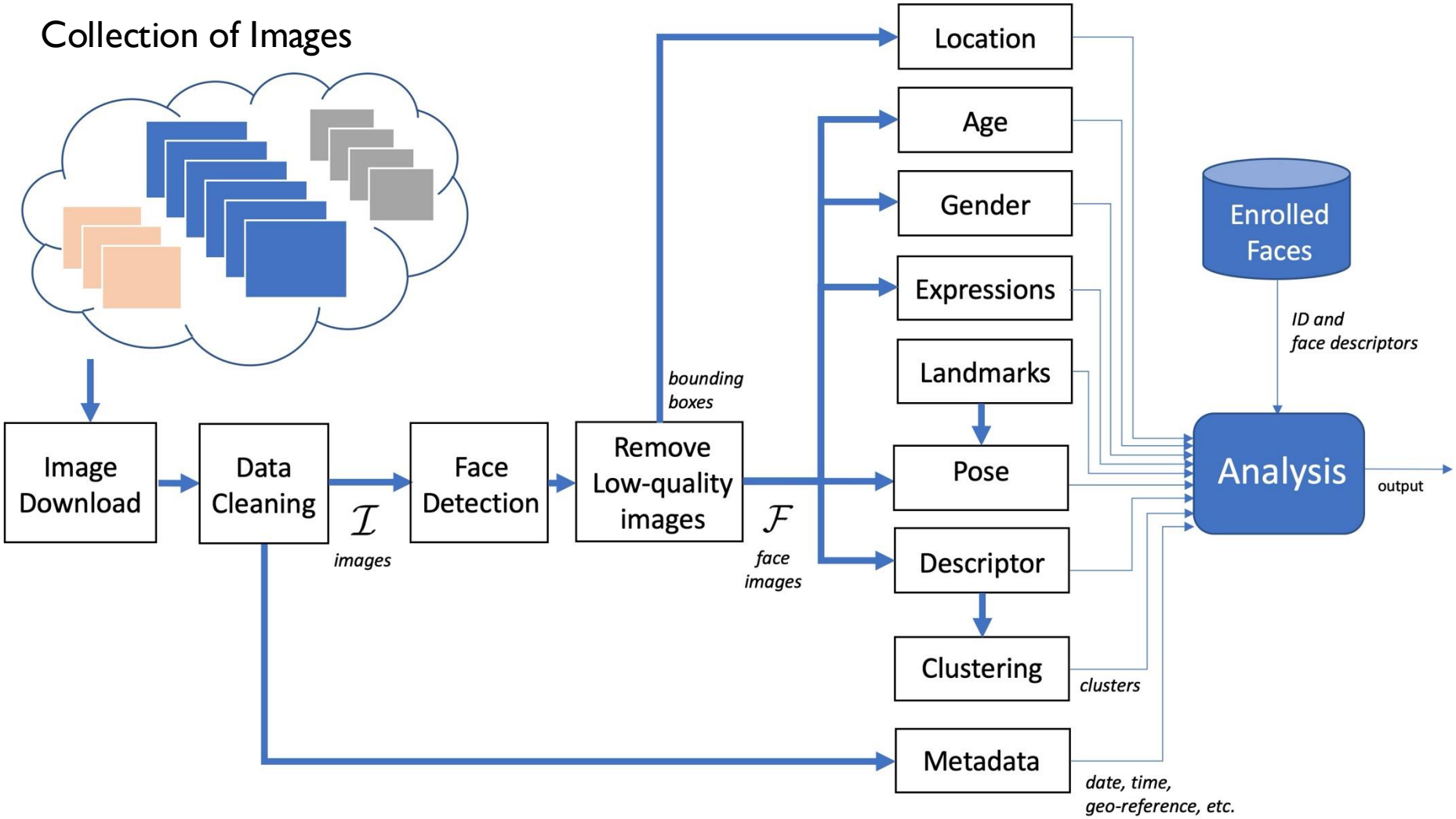
Ciencia de la Computación
Universidad Católica – Chile
http://domingomery.ing.uc.cl

(c) D.Mery - CV03D

La cara de una persona dice mucho…



Análisis Facial



1. Reconocimiento Facial

- La tarea es reconocer a una persona a través de su foto facial, o bien determinar si dos fotos de una cara pertenecen a la misma persona.

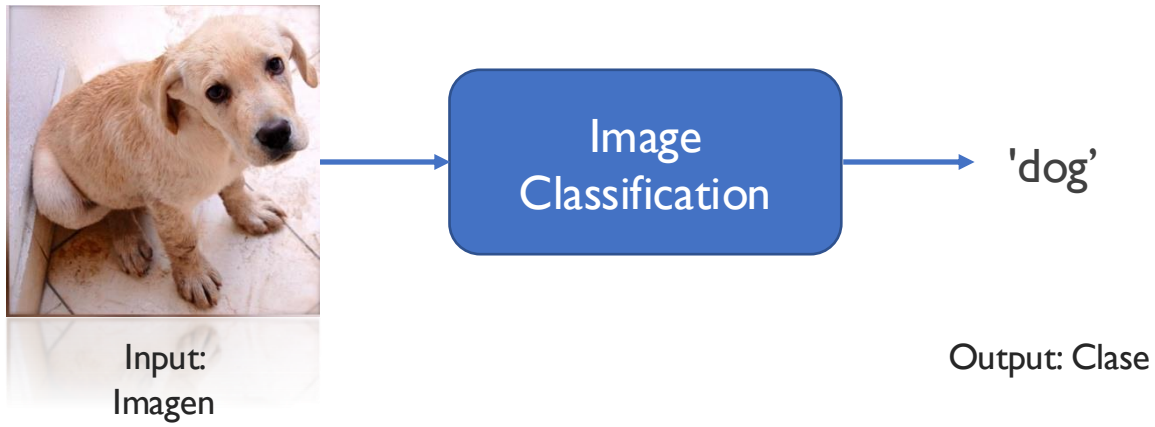


Foto Enrolada

Galería de Fotos

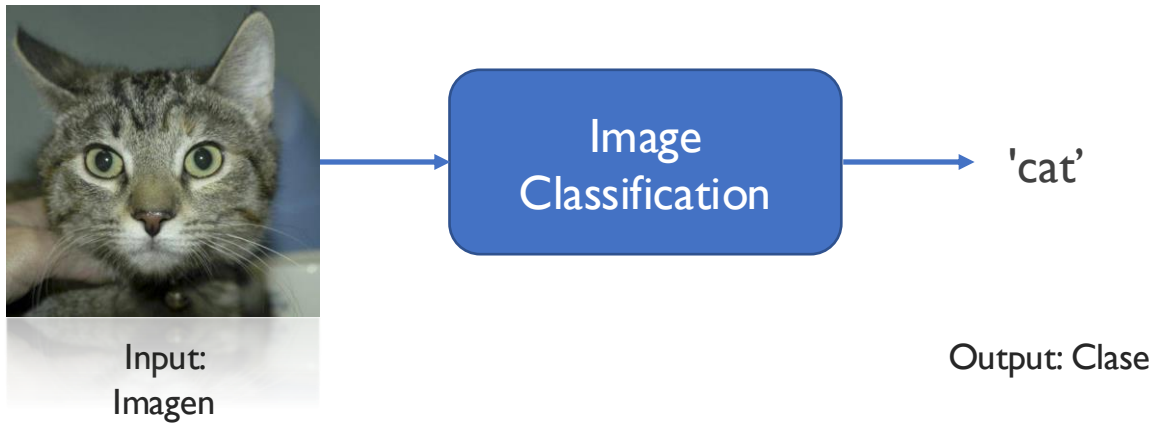
Clasificación de Imágenes – EL PROBLEMA

> Asignar una imagen a una clase o categoría



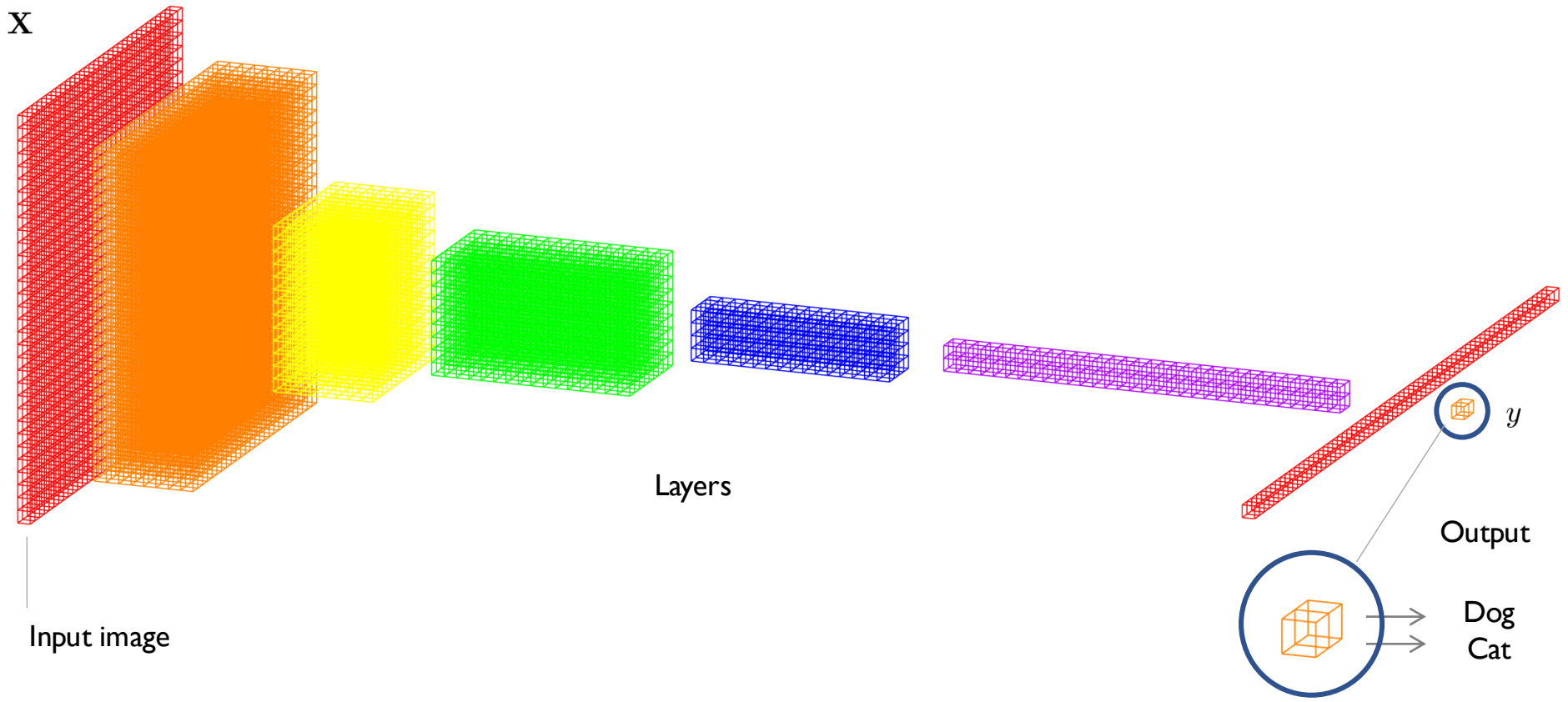
Clasificación de Imágenes – EL PROBLEMA

> Asignar una imagen a una clase o categoría



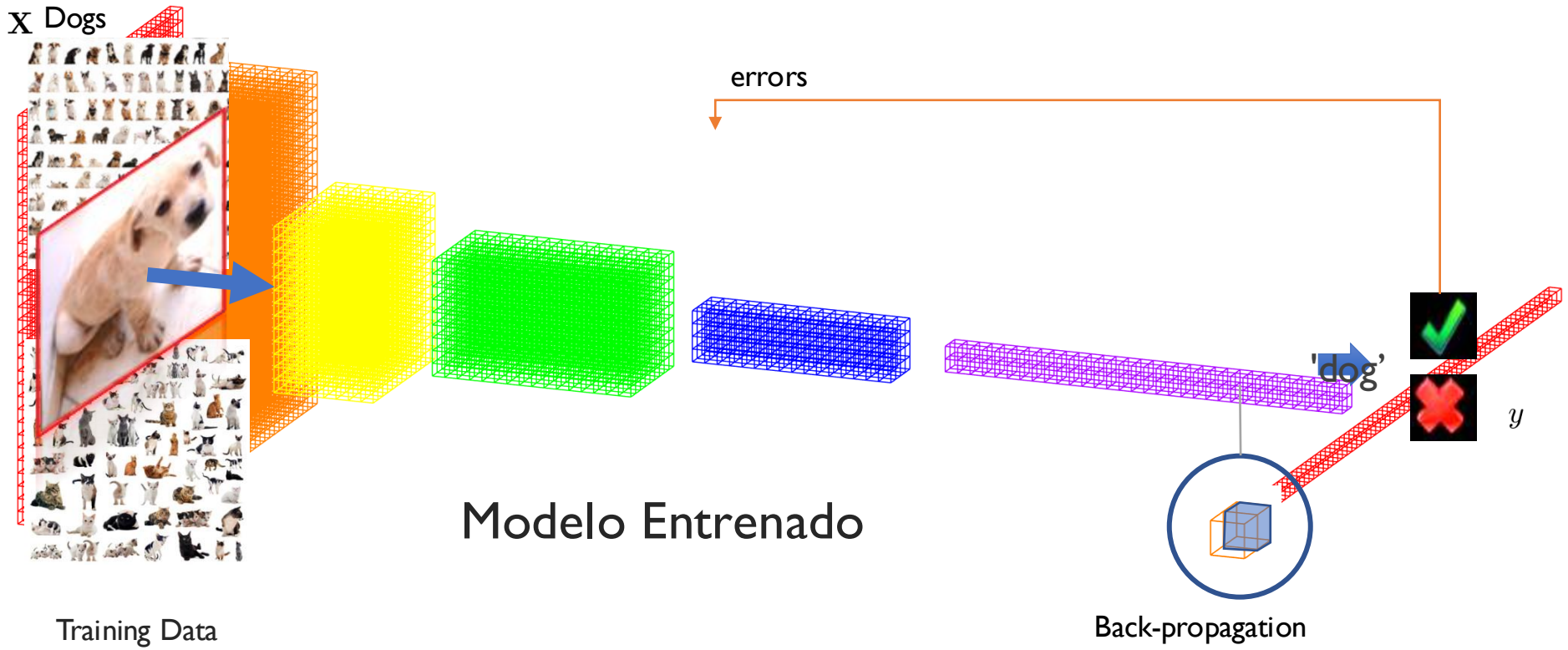
Clasificación de Imágenes – La Solución

Deep Learning



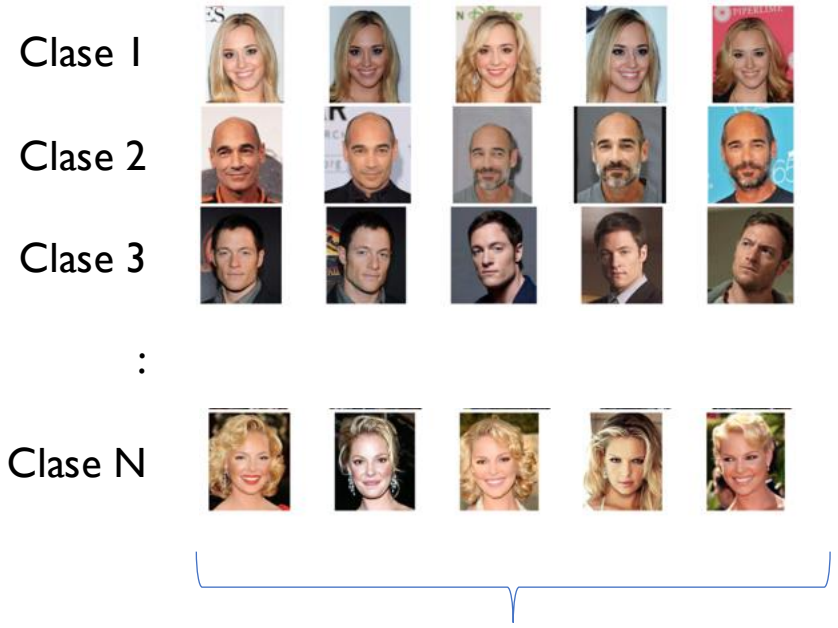
Clasificación de Imágenes – La Solución

Deep Learning

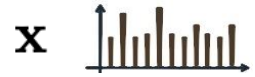


Reconocimiento Facial

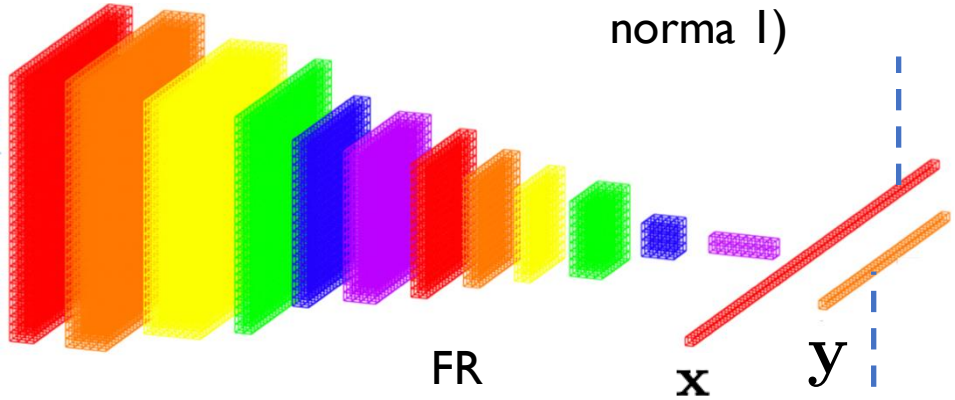
Reconocimiento Facial



N personas, cada una con más de 1000 fotos faciales

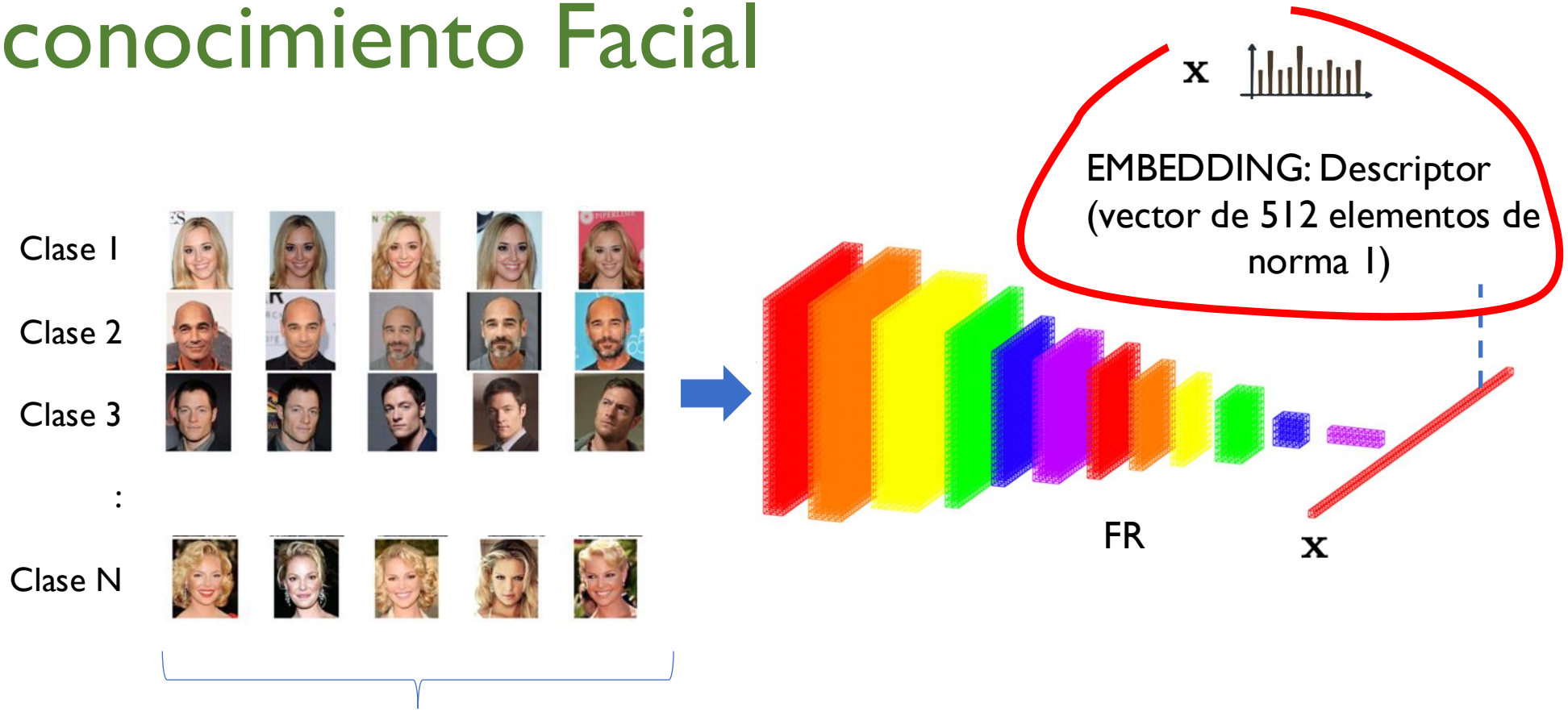


EMBEDDING: Descriptor (vector de 512 elementos de norma 1)



OUTPUT: Vector de probabilidades de N elementos

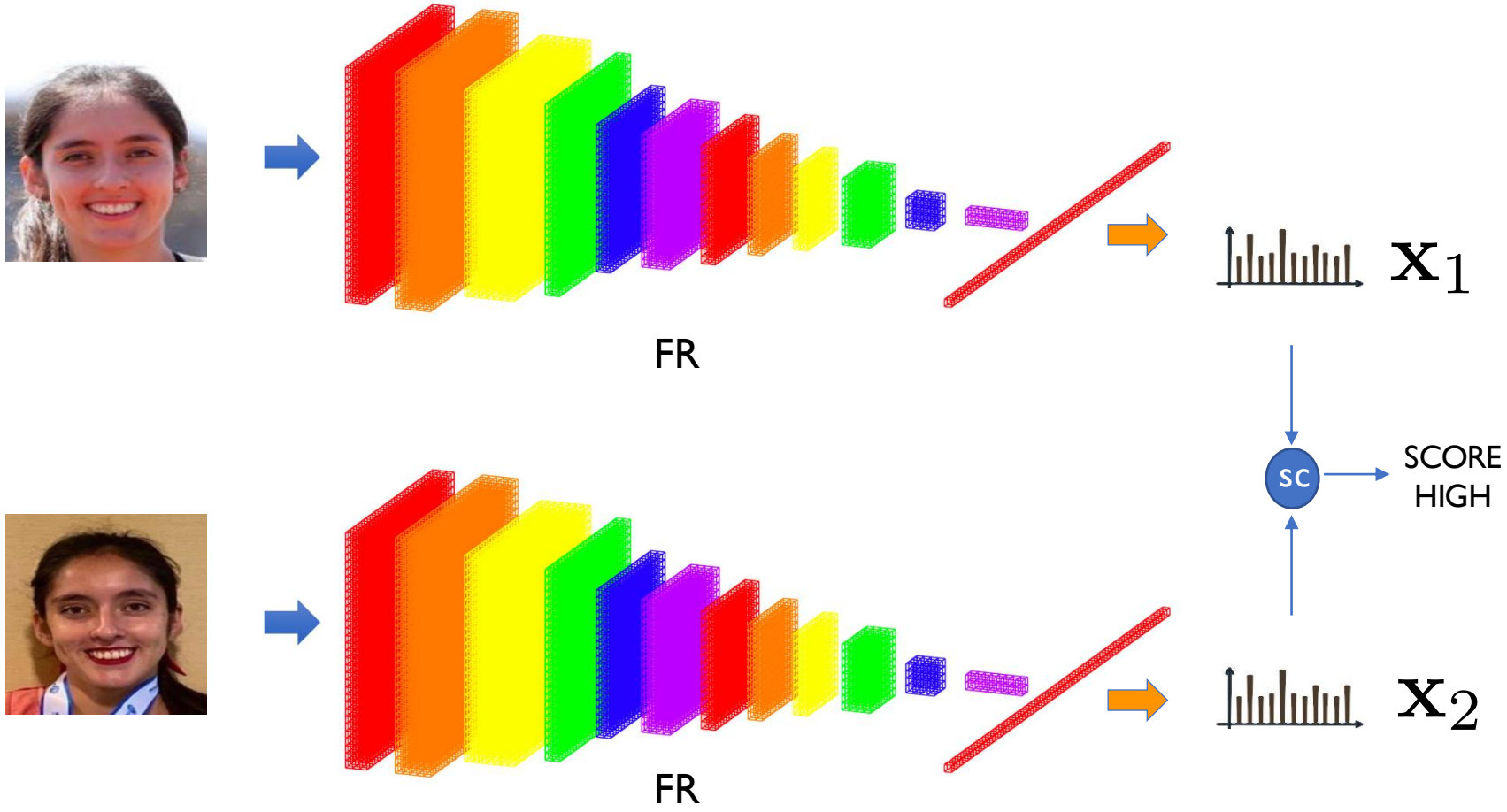
Reconocimiento Facial



Reconocimiento Facial

FOTOS DE LA MISMA PERSONA

SCORE:
mide qué tan parecidos
son los descriptores

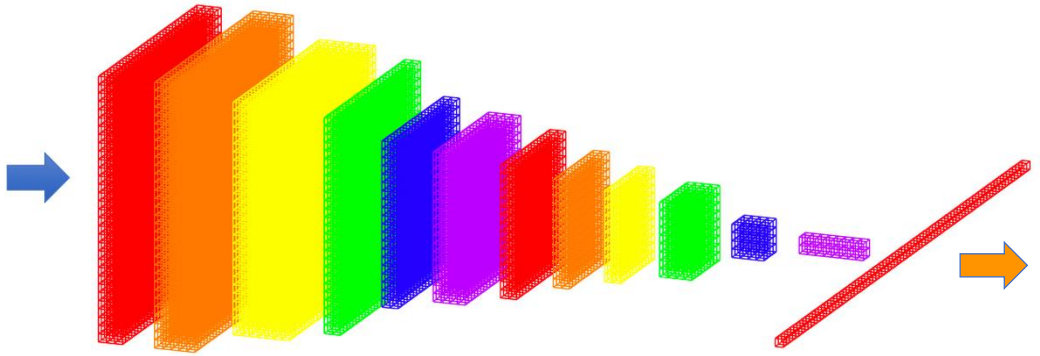


Reconocimiento Facial

FOTOS DE LA MISMA PERSONA

SCORE:
mide qué tan parecidos
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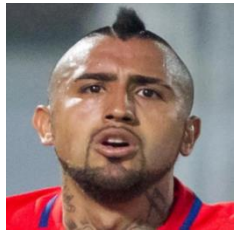


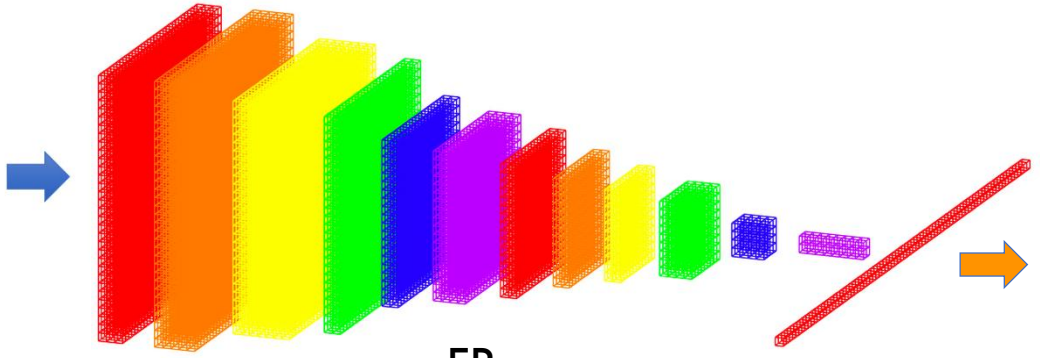
FR

$\mathbf{x}_1$

SC

SCORE
HIGH





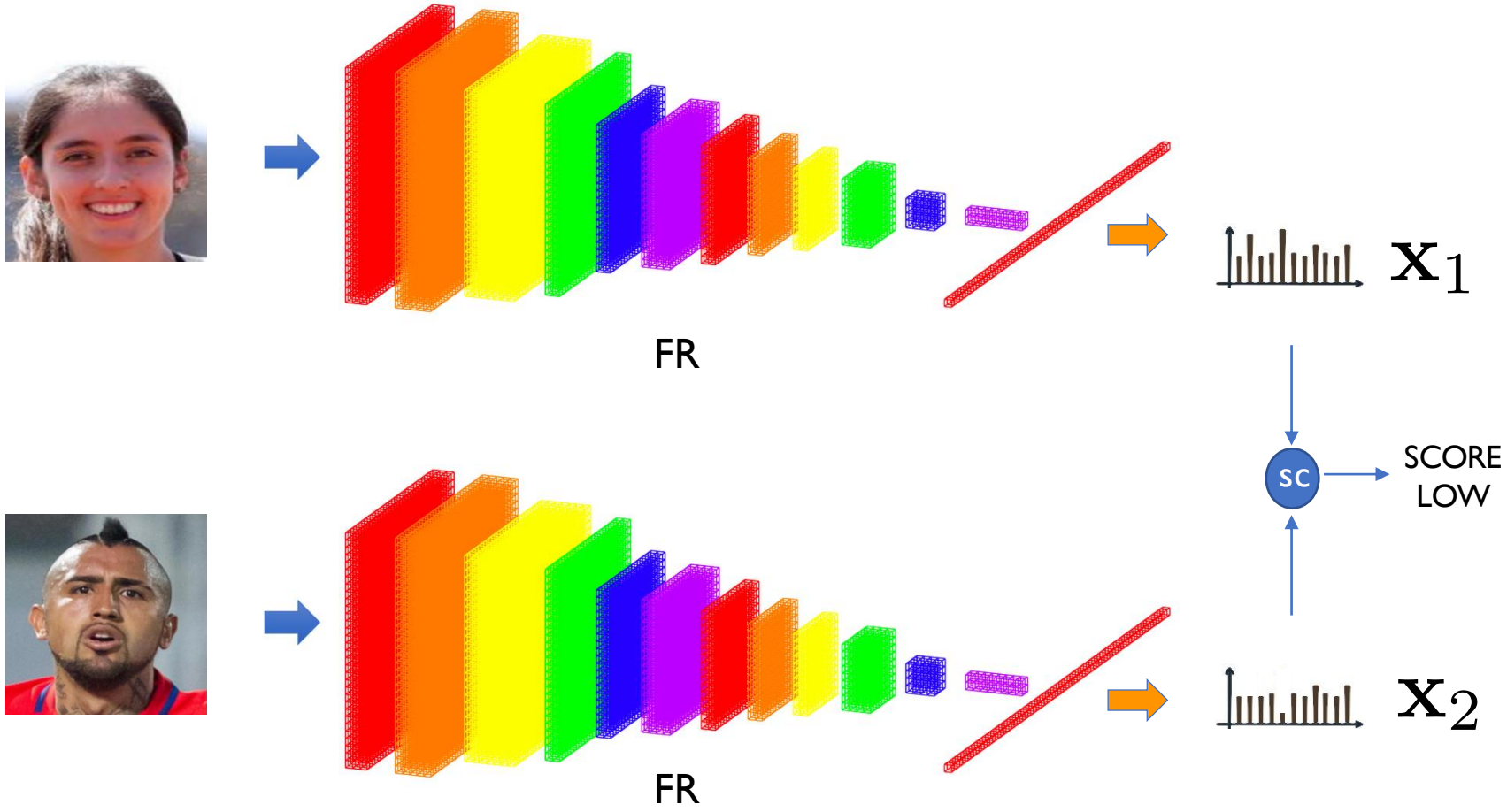
FR

$\mathbf{x}_2$

Reconocimiento Facial

FOTOS DE PERSONAS DISTINTAS

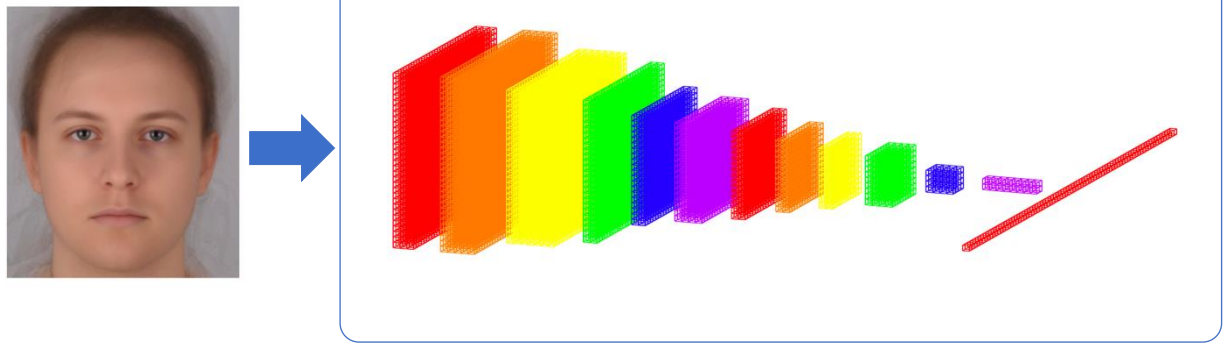
SCORE:
mide qué tan parecidos
son los descriptores



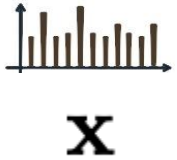
Reconocimiento Facial

Vector de 512 elementos
con norma 1

FR



DESCRIPTOR



$\mathbf{x}$

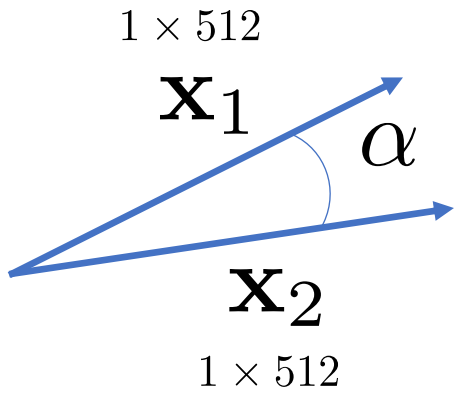
Input

Output

QUÉ TAN PARECIDOS SON DOS VECTORES?

NO OLVIDAR: en nuestro caso los vectores tienen 512 elementos y tienen norma 1

PRODUCTO PUNTO



$$\langle \mathbf{x}_1, \mathbf{x}_2 \rangle = ||\mathbf{x}_1|| ||\mathbf{x}_2|| \cos(\alpha)$$

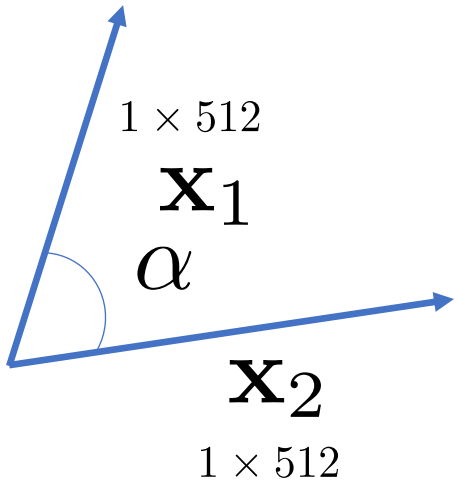
$$\langle \mathbf{x}_1, \mathbf{x}_2 \rangle = \cos(\alpha) \quad \text{SCORE}$$

$$\langle \mathbf{x}_1, \mathbf{x}_2 \rangle = \mathbf{x}_1 \mathbf{x}_2^{\mathsf{T}}$$

- Si los vectores son muy parecidos, el ángulo entre ellos es pequeño y el SCORE es cercano a 1.

QUÉ TAN PARECIDOS SON DOS VECTORES?

NO OLVIDAR: en nuestro caso los vectores tienen 512 elementos y tienen norma 1



PRODUCTO PUNTO

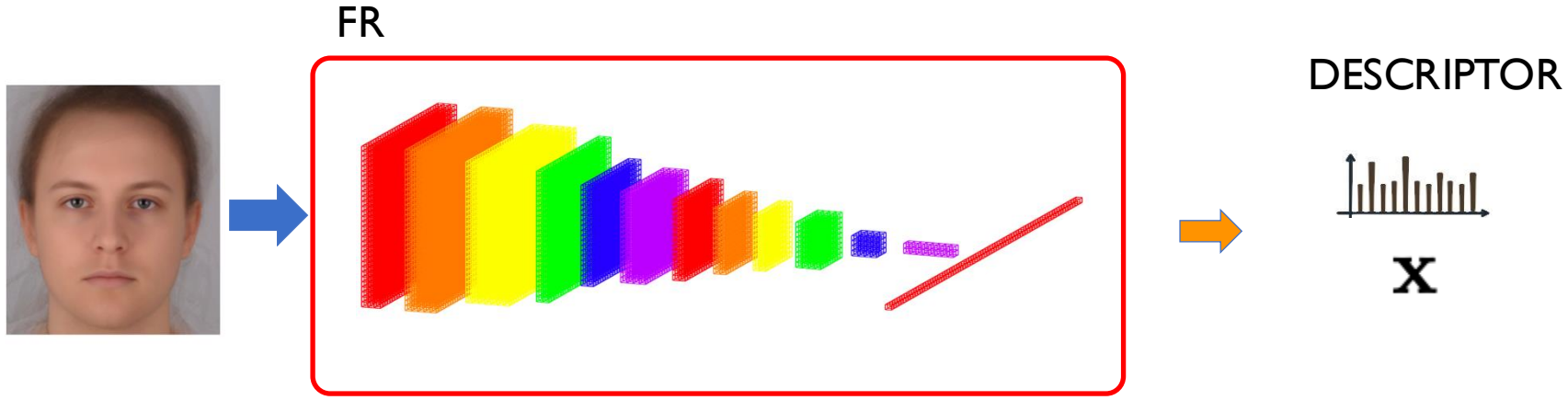
$$\langle \mathbf{x}_1, \mathbf{x}_2 \rangle = ||\mathbf{x}_1|| ||\mathbf{x}_2|| \cos(\alpha)$$

$$\langle \mathbf{x}_1, \mathbf{x}_2 \rangle = \cos(\alpha)$$ SCORE

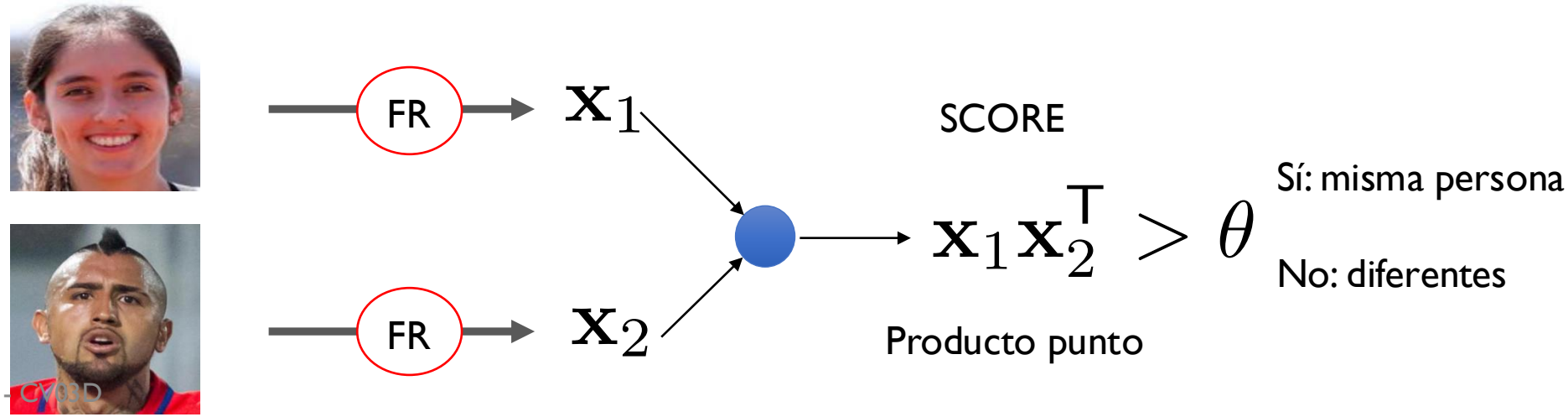
$$\langle \mathbf{x}_1, \mathbf{x}_2 \rangle = \mathbf{x}_1 \mathbf{x}_2^{\mathsf{T}}$$

- Si los vectores son muy parecidos, el ángulo entre ellos es pequeño y el SCORE es cercano a 1.
- Si los vectores son muy distintos, el ángulo entre ellos es grande y el SCORE disminuye significativamente.

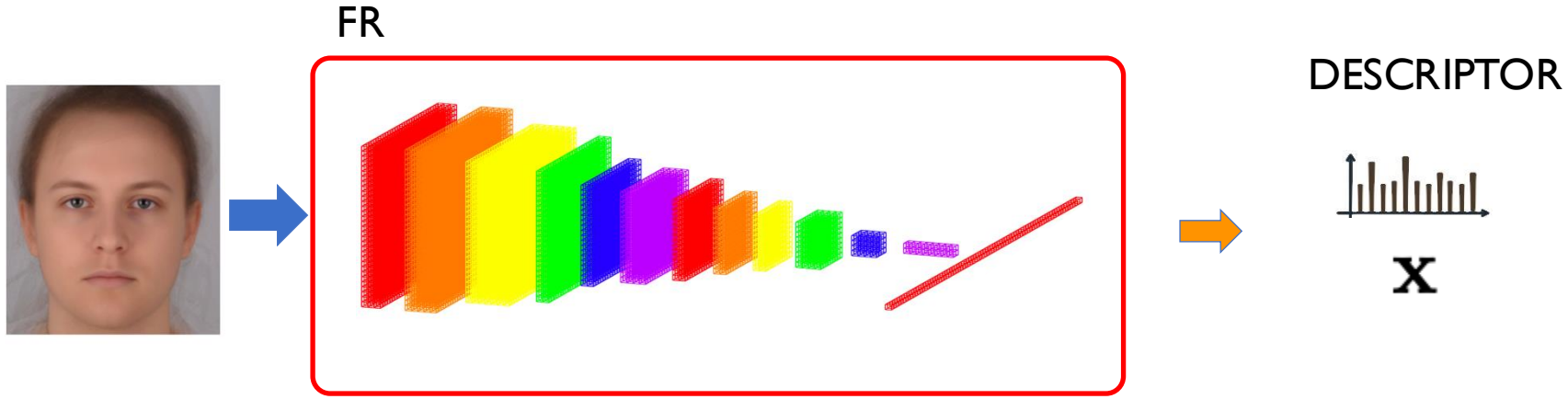
Reconocimiento Facial



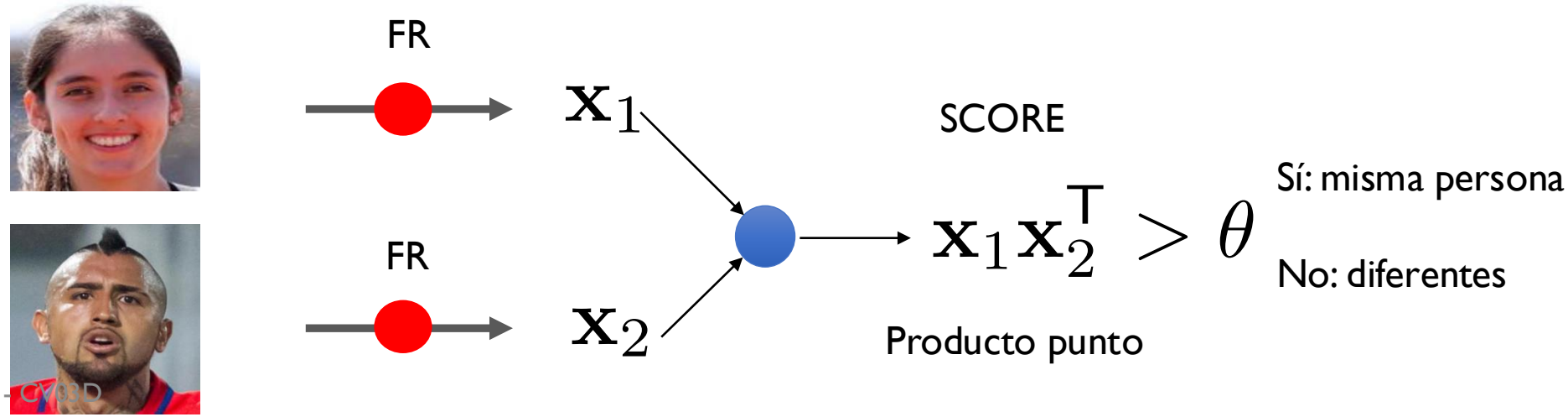
Fotos faciales de la misma (diferente) persona tienen descriptores similares (diferentes)



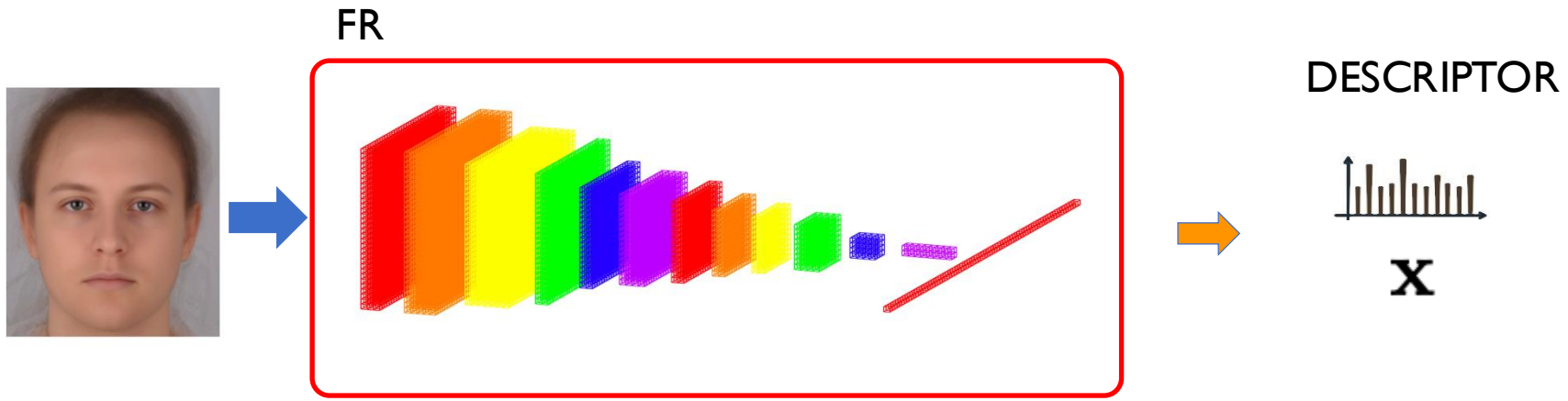
Reconocimiento Facial



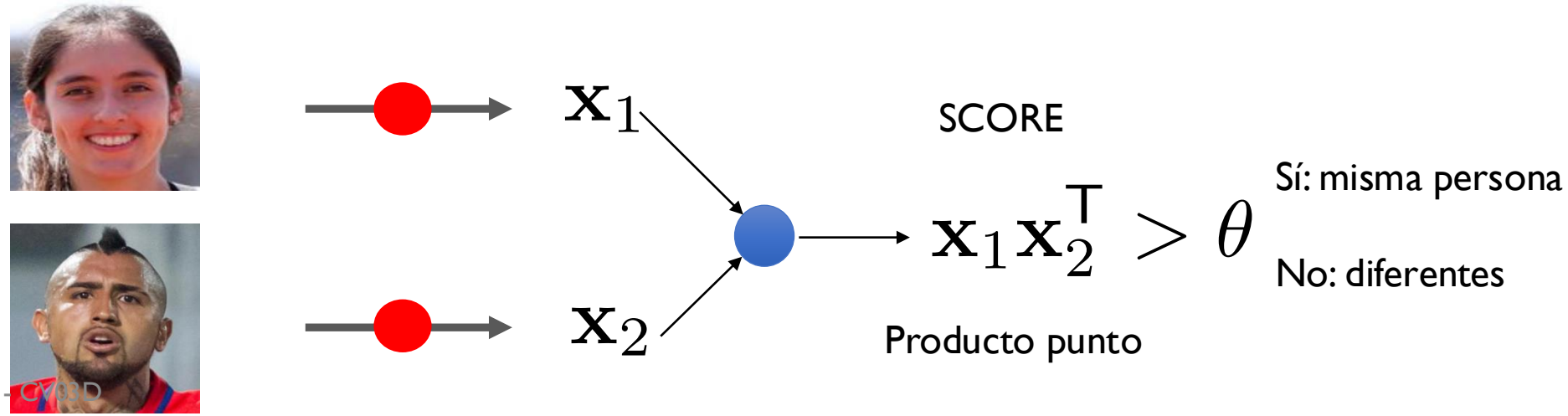
Fotos faciales de la misma (diferente) persona tienen descriptores similares (diferentes)



Reconocimiento Facial

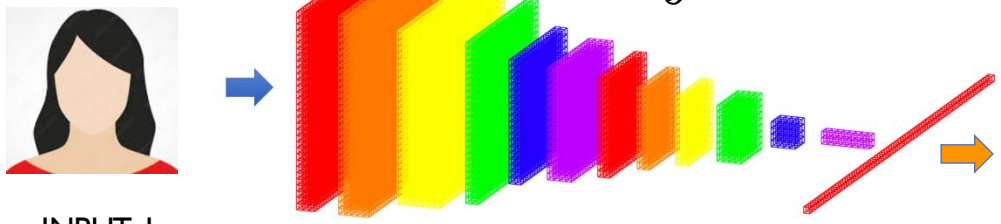


Fotos faciales de la misma (diferente) persona tienen descriptores similares (diferentes)



Embeddings

EMBEDDDINGS: vectores de n elementos que describen la entrada. Normalmente, tienen una norma igual a 1 y n=512.



INPUT 1

$\mathbf{a}_1$

f

EMBEDDING 1

$\mathbf{x}_1 = f(\mathbf{a}_1)$

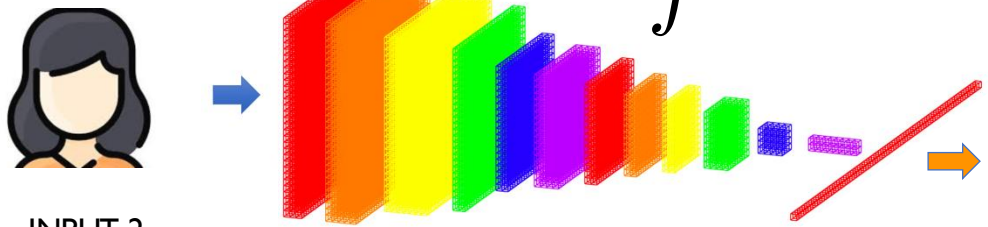
sc

$sc(\mathbf{x}_1, \mathbf{x}_2)$

SCORE: mide que tan similares son los embeddings

HIGH → similar

LOW → diferentes



INPUT 2

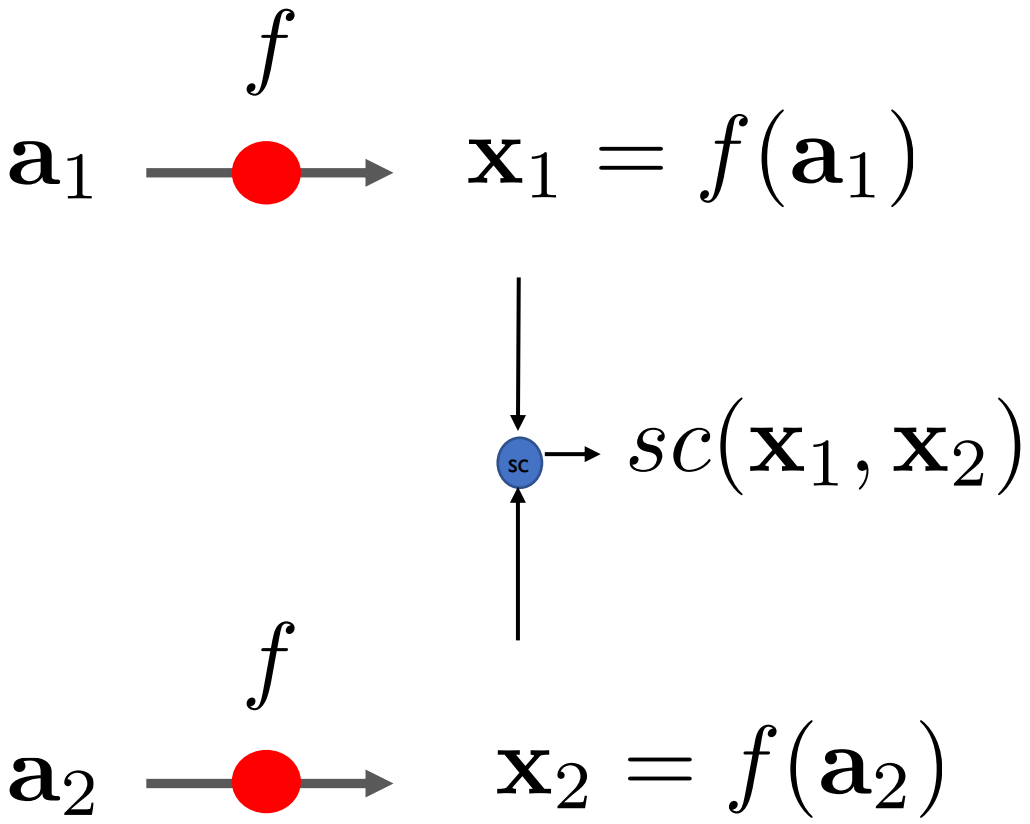
$\mathbf{a}_2$

f

EMBEDDING 2

$\mathbf{x}_2 = f(\mathbf{a}_2)$

Embeddings


f
$\mathbf{a}_1$
$\mathbf{x}_1 = f(\mathbf{a}_1)$
sc
$sc(\mathbf{x}_1, \mathbf{x}_2)$
f
$\mathbf{a}_2$
$\mathbf{x}_2 = f(\mathbf{a}_2)$

Ejemplo: Quién es esta persona?



DESCRIPTOR

$\mathbf{x}$

1×512

Ejemplo: Quién es esta persona?



DESCRIPTOR

$\mathbf{x}^T$

512×1

BASE DE DATOS:



$\mathbf{y}_1$



$\mathbf{y}_2$









$\mathbf{y}_m$

$\mathbf{Y}$

$m \times 512$

$\cdot$

$=$

$\mathbf{s}$

$m \times 1$

Ejemplo: Quién es esta persona?

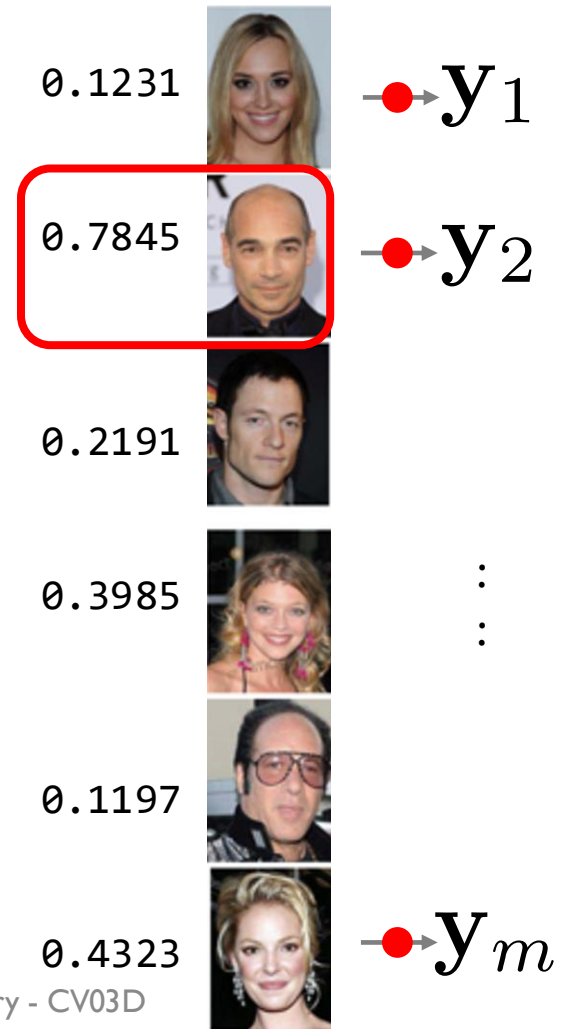


DESCRIPTOR

$\mathbf{x}^T$

512×1

BASE DE DATOS:



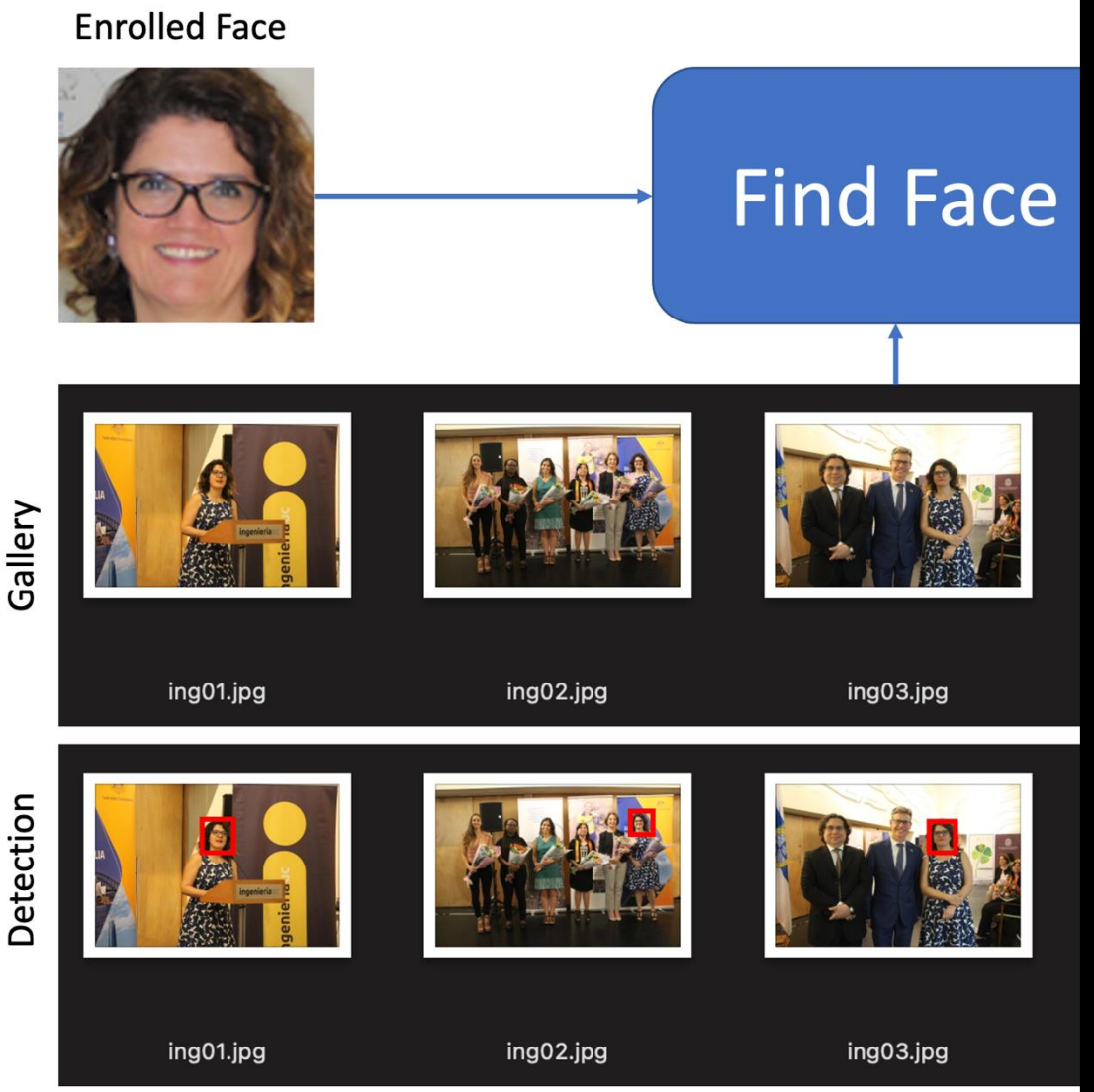
Producto punto del descriptor de prueba con todos los descriptores de la base de datos:

$$\underset{m \times 1}{\mathbf{s}} = \mathbf{Y}\mathbf{x}^T \quad (s_j = \mathbf{y_j} \cdot \mathbf{x})$$

Se busca en $\mathbf{s}$ aquellos valores mayores a un threshold. De esta manera, la persona j de la base de datos hace matching con la persona de prueba, si se cumple:

$$s_j > \theta$$

Ejemplo: Búsqueda en una

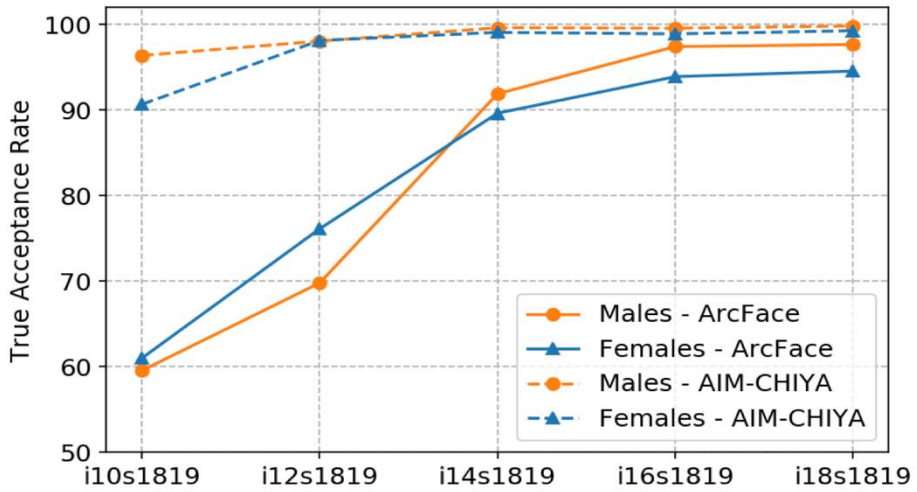


PASOS:

1. Descriptor de cara enrolada (x)
2. Detectar las caras de la galería
3. Por cada cara detectada extraer descriptor (y_j), y guardar coordenadas de cada bounding box.
4. Generar matriz Y, y calcular
   $s = Yx^T$
5. Buscar los elementos de s mayores a umbral
6. Marcar las caras que cumplan condición anterior

1. Reconocimiento Facial

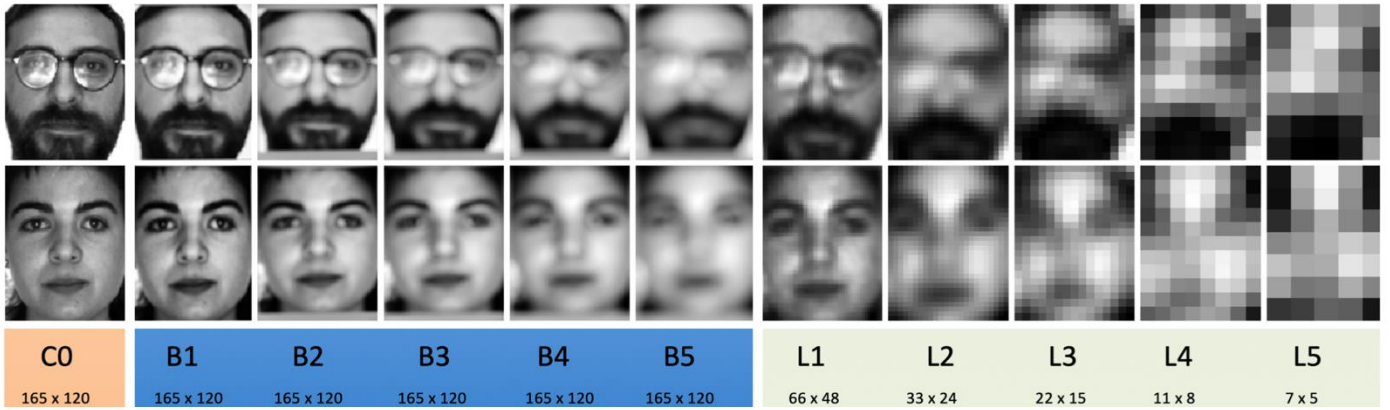


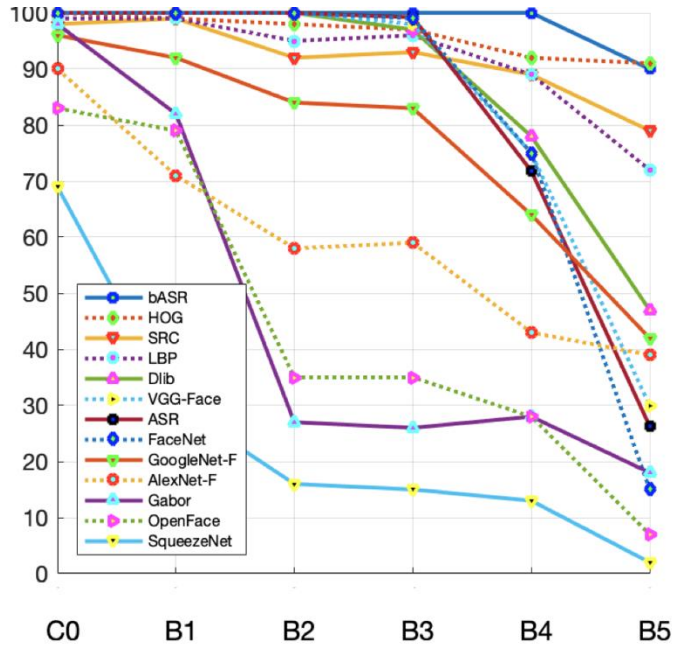


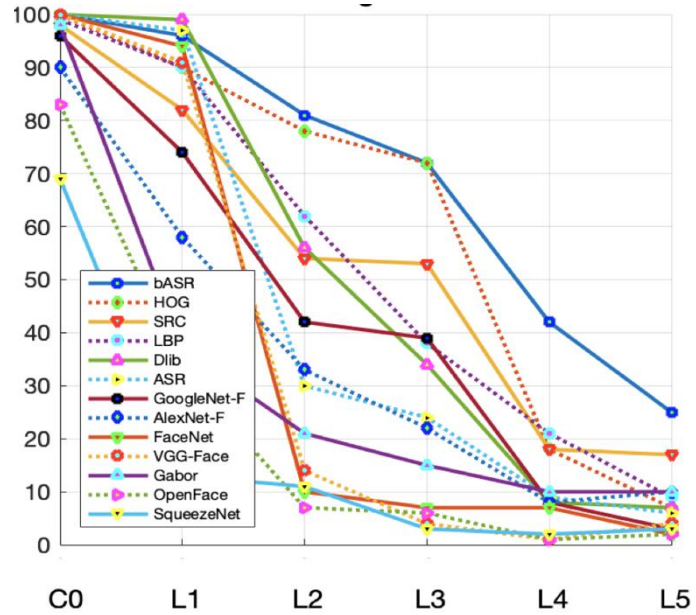
(a) False Acceptance Rate = 0.01%

Identity Document to Selfie Face Matching Across Adolescence, Albiero, V.; Srinivas, N.; Villalobos, E.; Perez-Facuse, J.; Rosenthal, R.; Mery, D.; Ricanek, K.; Bowyer. K., International Joint Conference on Biometrics (IJCB 2020).

1. Reconocimiento Facial







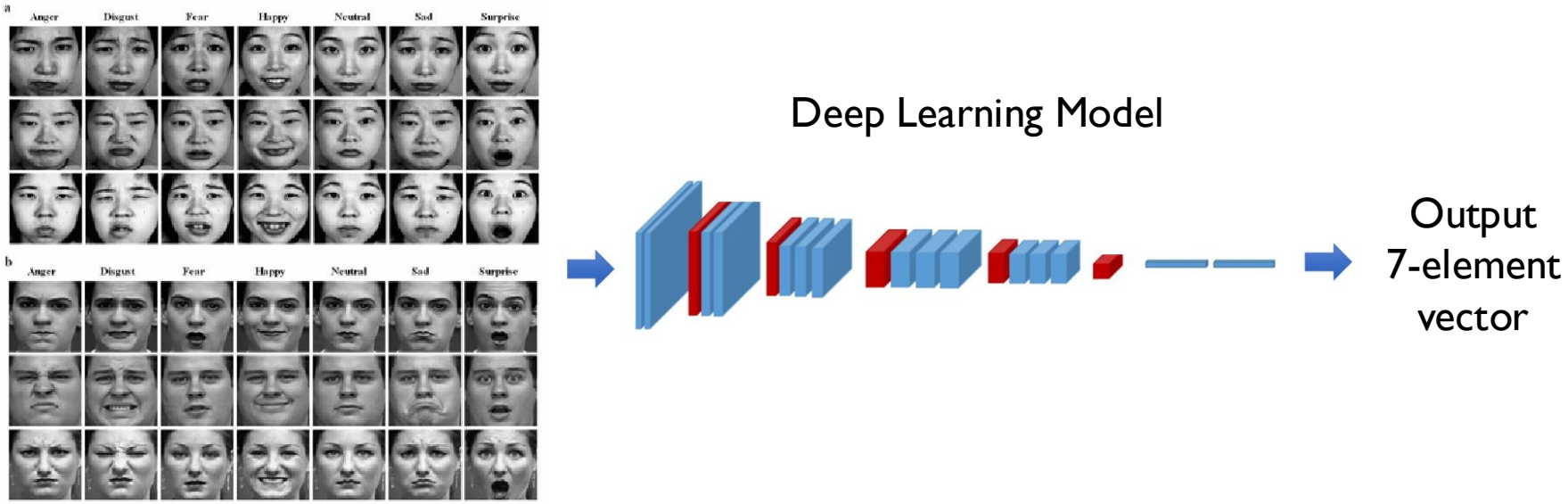
Face Recognition in Low-Quality Images using Adaptive Sparse Representations, Heinsohn, D.; Villalobos, E.; Prieto, L.; and Mery, D. Image and Vision Computing. 2019

2. Expresiones Faciales



Dataset:
Faces with expressions
7-class problem

2. Expresiones Faciales



Angry	Disgust	Scared	Happy	Sad	Surprised	Neutral
0.2161	0.2403	0.0337	0.2423	0.16780	0.0259	0.0739

$$\sum = 1$$

3. Reconocimiento de Género

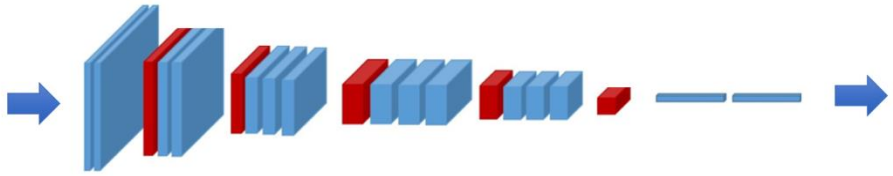


Dataset:
Female and Male Faces
2-Class Problem

3. Reconocimiento de Género



Deep Learning Model

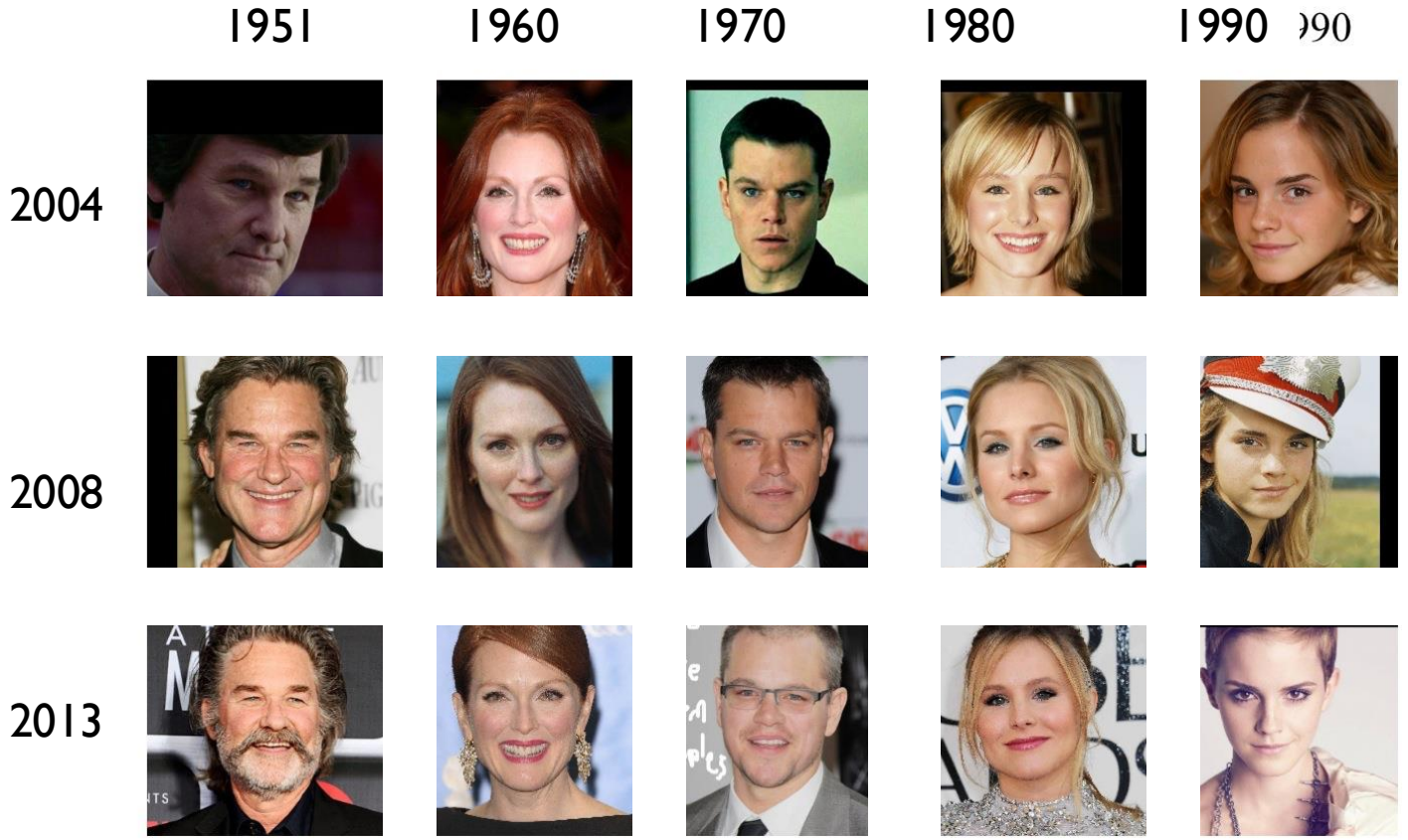


Output
2-element
vector

Male	Female
0.74	0.26

$$\sum = 1$$

4. Reconocimiento de Edad

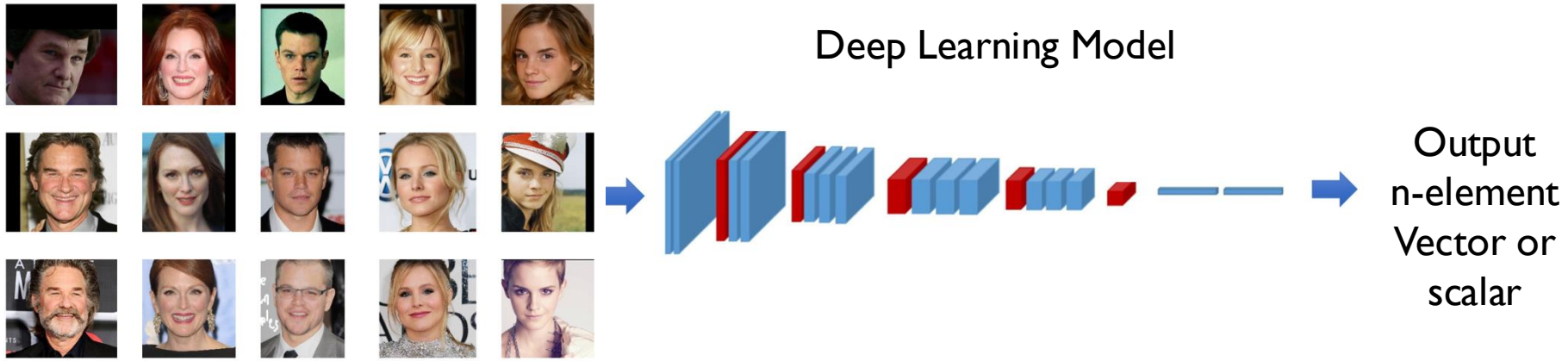


Dataset:
Faces with age

Two approaches:

- Output is a class (class 1: from 0 to 2 years, class 2: from 3 to 8, etc.)
- Output is a scalar (age of the person)

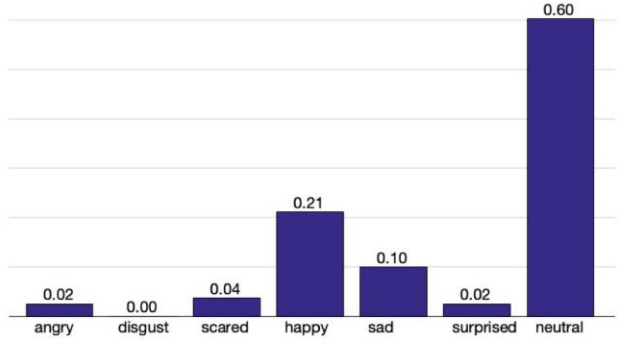
4. Reconocimiento de Edad



Example

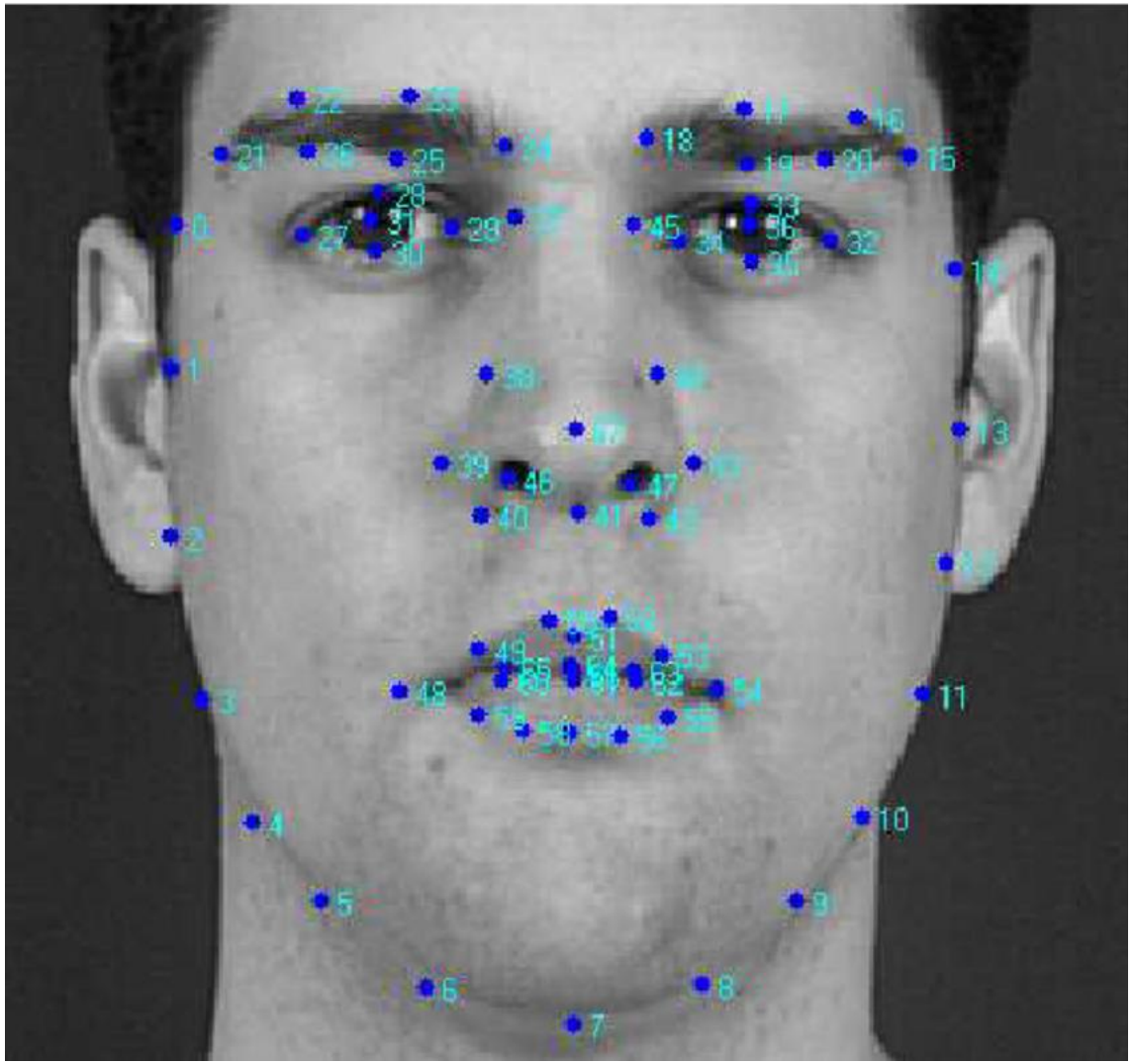

1
2
3
4
5
6
7
8
9
10
11
12


1 2 3 4 5 6 7 8 9 10 11 12

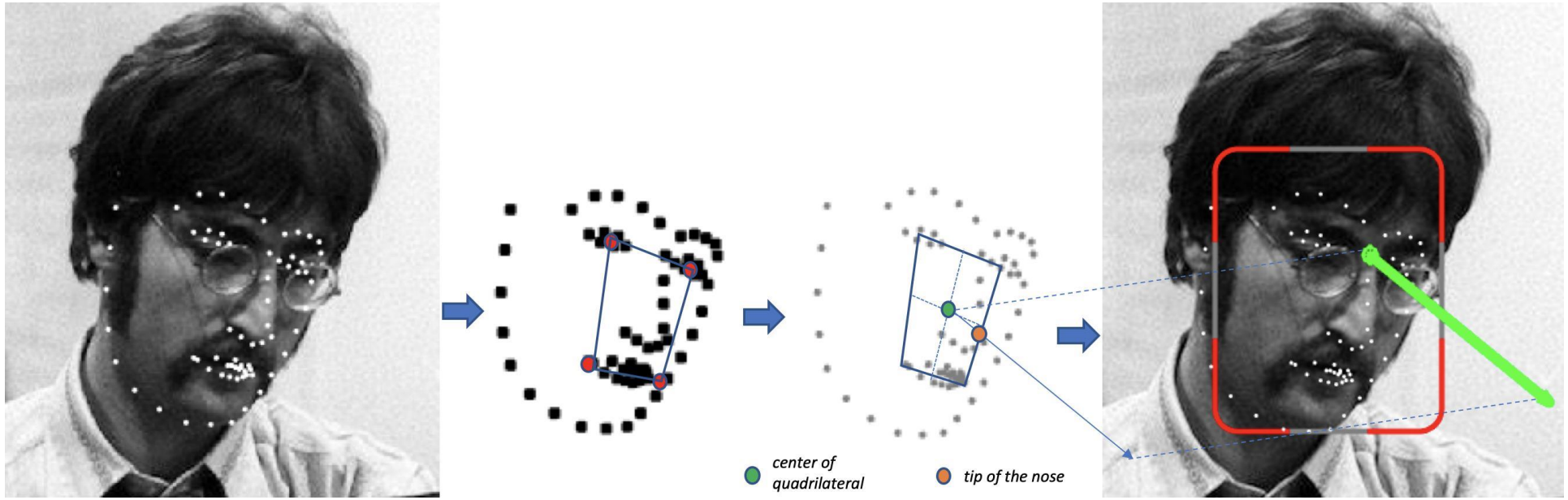

0.02
0.00
0.04
0.21
0.10
0.02
0.60
angry
disgust
scared
happy
sad
surprised
neutral



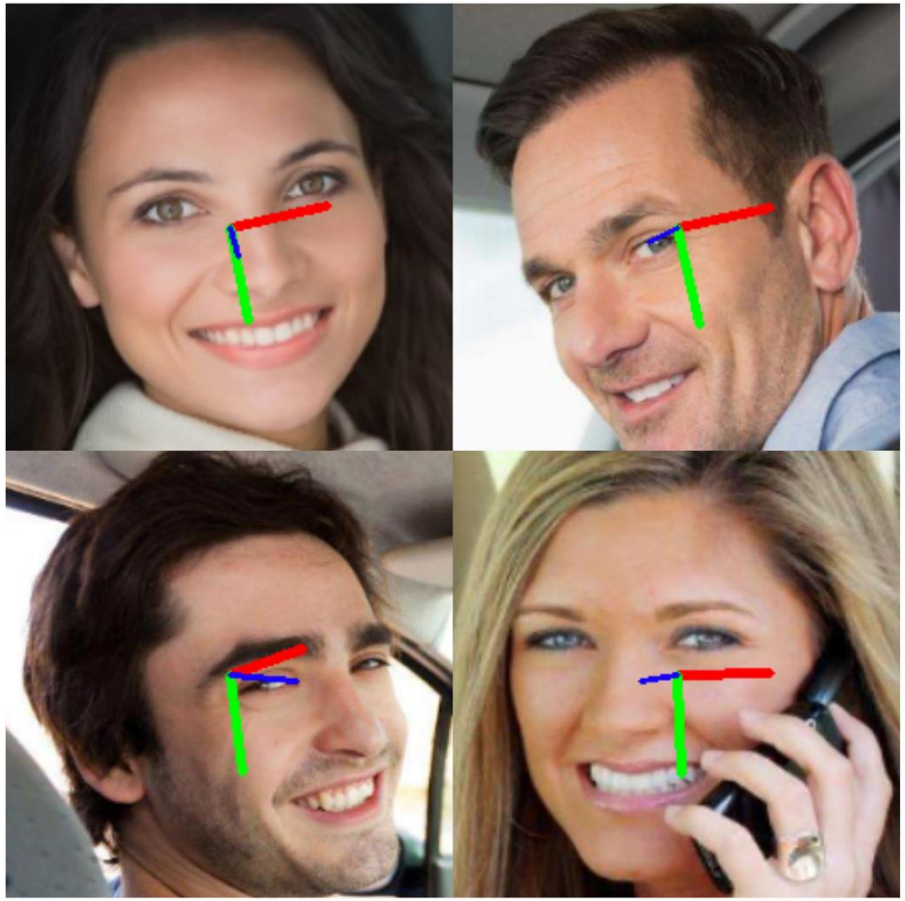
5. Landmarks

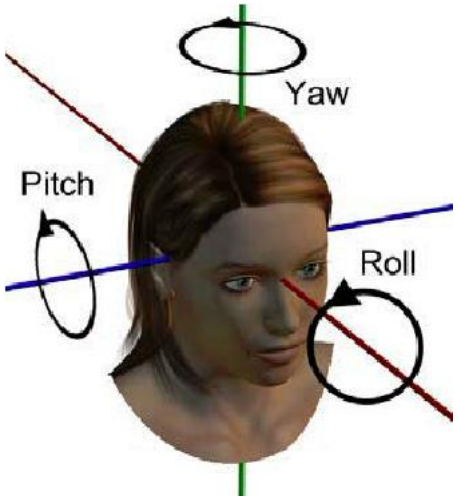


6. Estimación de Pose (from landmarks)



6. Estimación de Pose



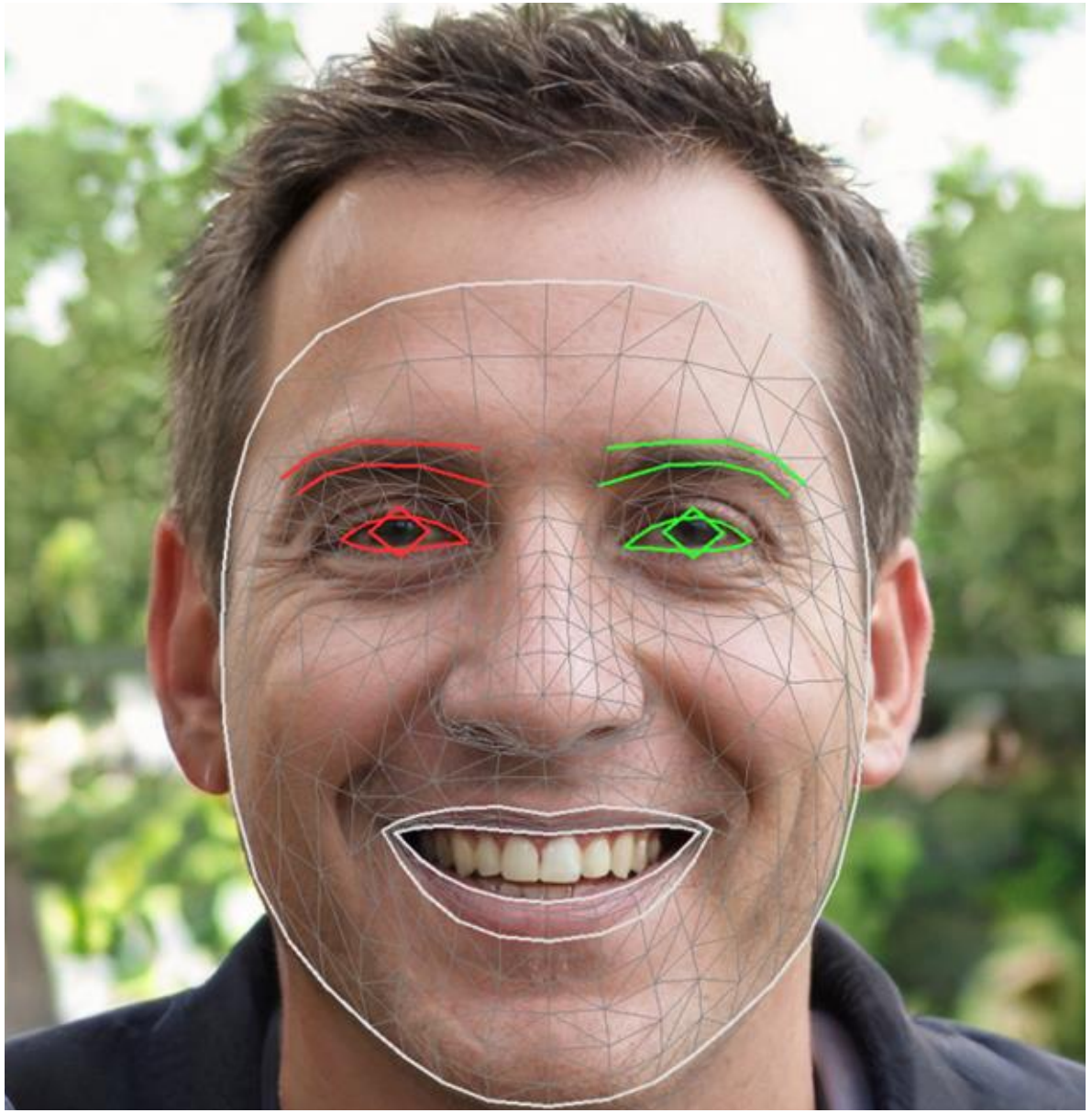


Yaw: ‘say no’
Pitch: ‘say yes’
Roll: ‘say maybe’

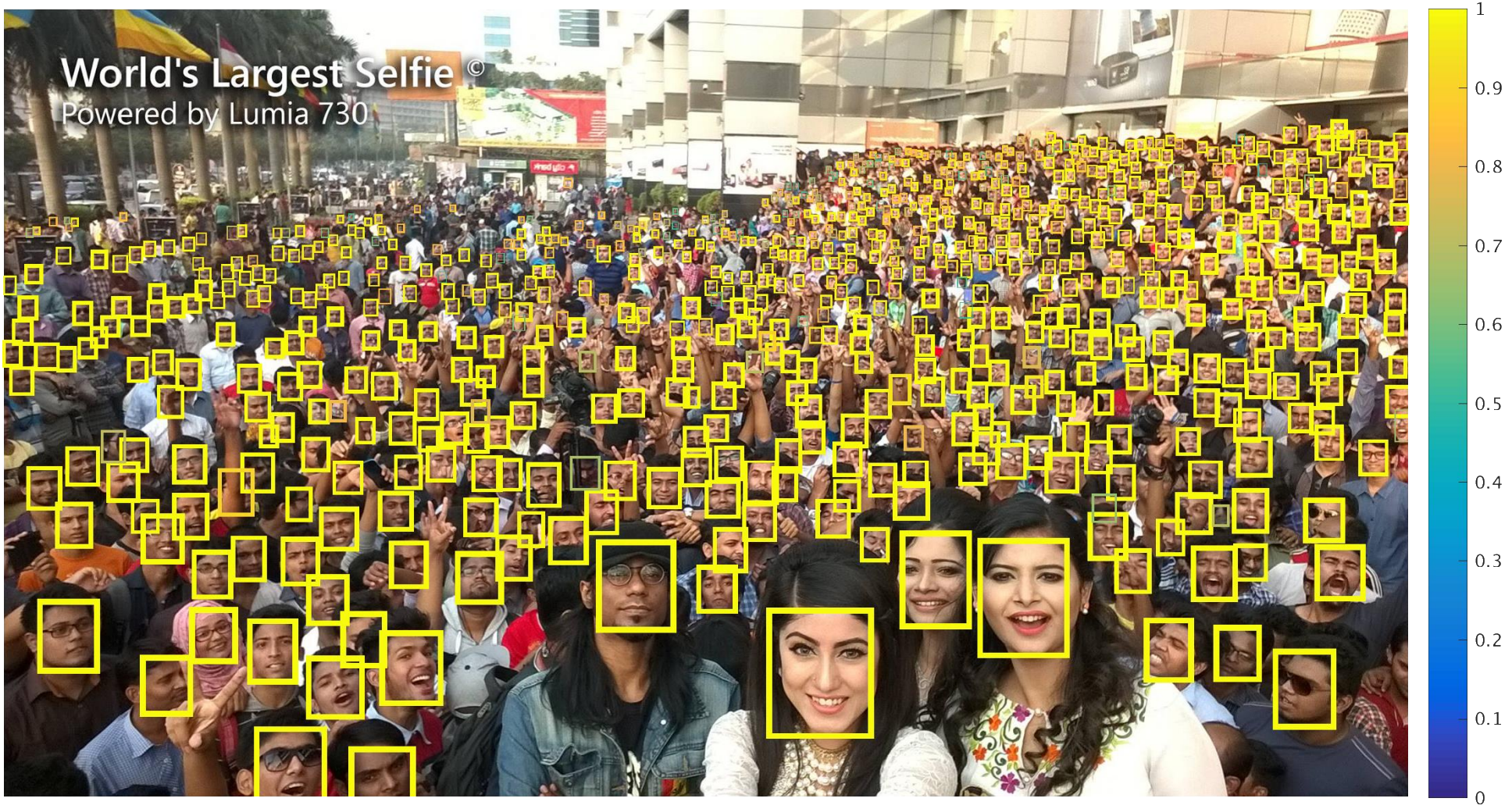
7. Mesh



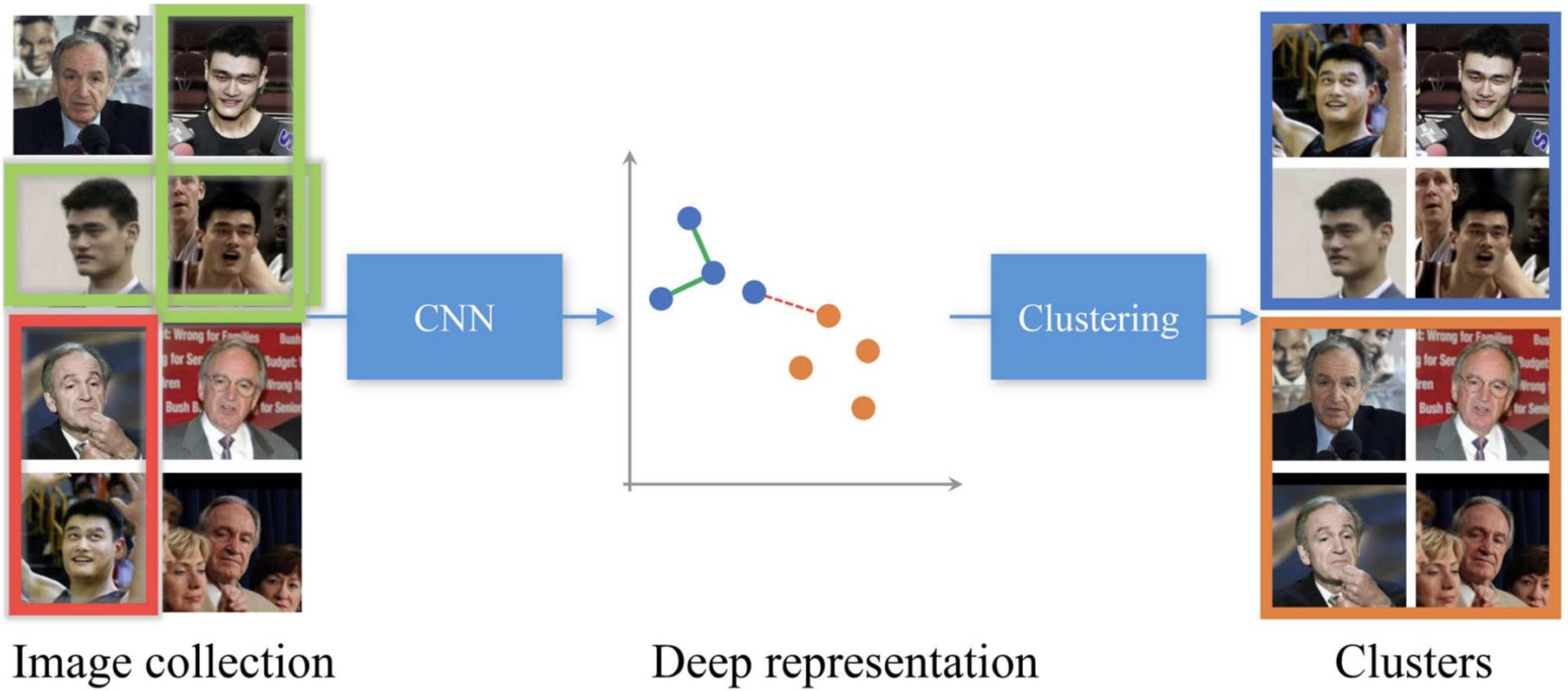
7. Mesh



8. Detección de Caras



9. Clustering de Caras



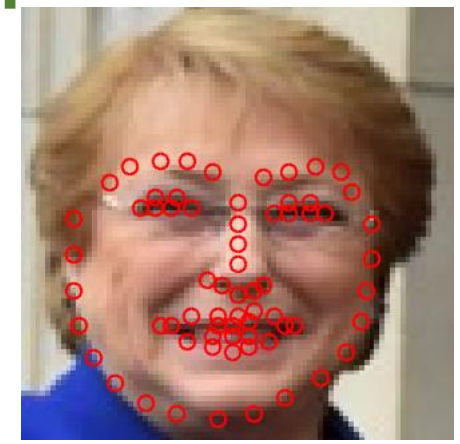
Face Clustering



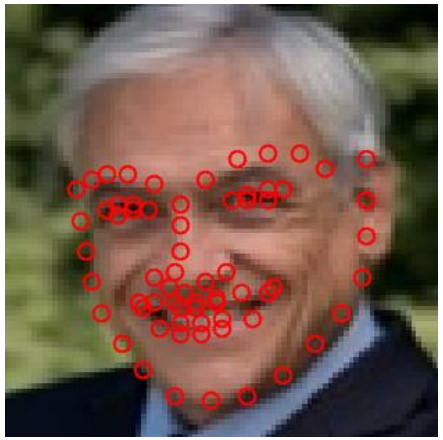
Face Clustering



Ejemplo



Gender	Age	Angry	Disgust	Scared	Happy	Sad	Surprised	Neutral
0.82	58.5	0.0174	0.0005	0.0045	0.8540	0.0692	0.0011	0.0532



Gender	Age	Angry	Disgust	Scared	Happy	Sad	Surprised	Neutral
0.04	59.9	0.0012	0.0001	0.0014	0.9371	0.0109	0.0001	0.0491

Co-occurrence

- Images in which two persons are present



Co-occurrence



Co-occurrence

- Images in which two persons are present

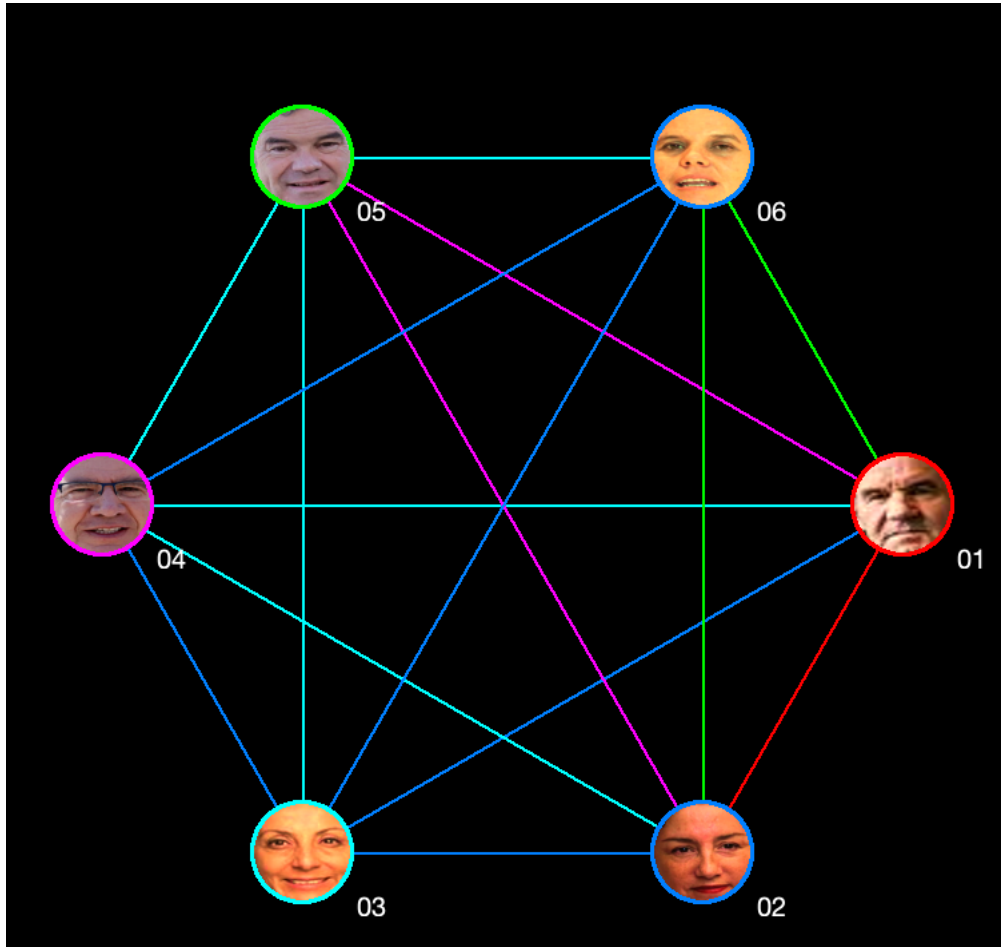
	S1	S2	S3	S4	S5	S6
S1	-	12	4	5	6	3
S2	12	-	3	4	1	2
S3	4	3	-	8	5	12
S4	5	4	8	-	3	1
S5	6	1	5	3	-	5
S6	3	2	12	1	5	-

Co-occurrence Matrix for 6 subjects

Subjects 1 and 2 are present in 12 pictures

Co-occurrence

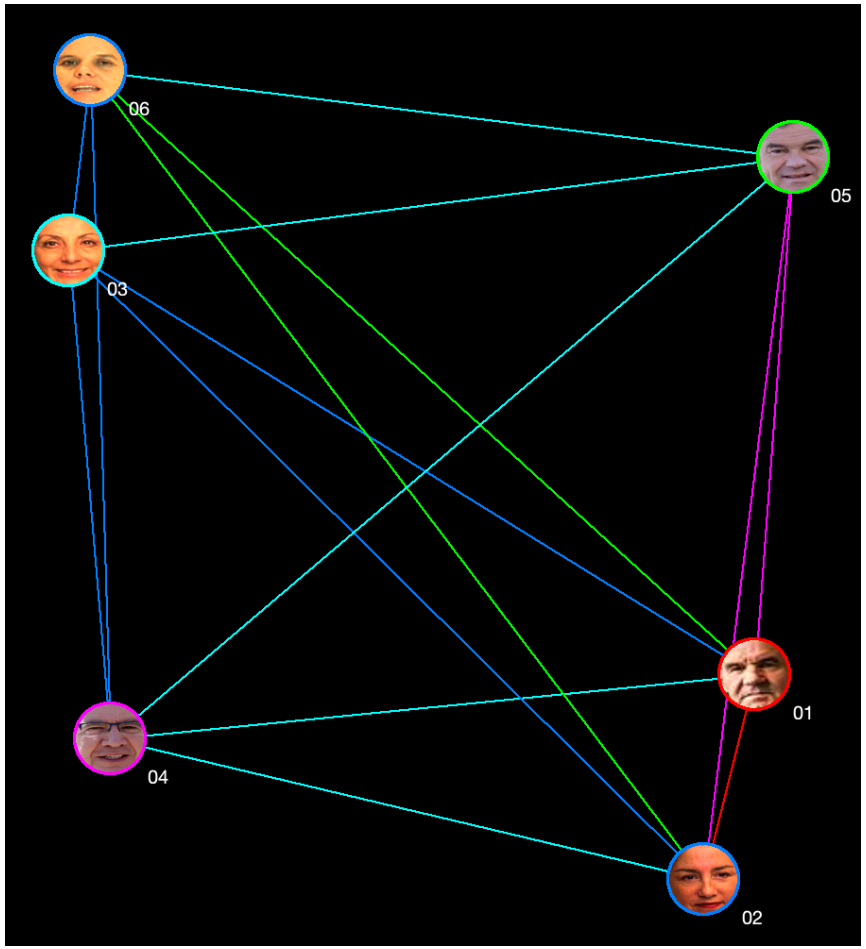
	S1	S2	S3	S4	S5	S6
S1	-	12	4	5	6	3
S2	12	-	3	4	1	2
S3	4	3	-	8	5	12
S4	5	4	8	-	3	1
S5	6	1	5	3	-	5
S6	3	2	12	1	5	-



Co-occurrence

	S1	S2	S3	S4	S5	S6
S1	-	12	4	5	6	3
S2	12	-	3	4	1	2
S3	4	3	-	8	5	12
S4	5	4	8	-	3	1
S5	6	1	5	3	-	5
S6	3	2	12	1	5	-

New representation:
Closeness mean more co-occurrence



Graph Representation

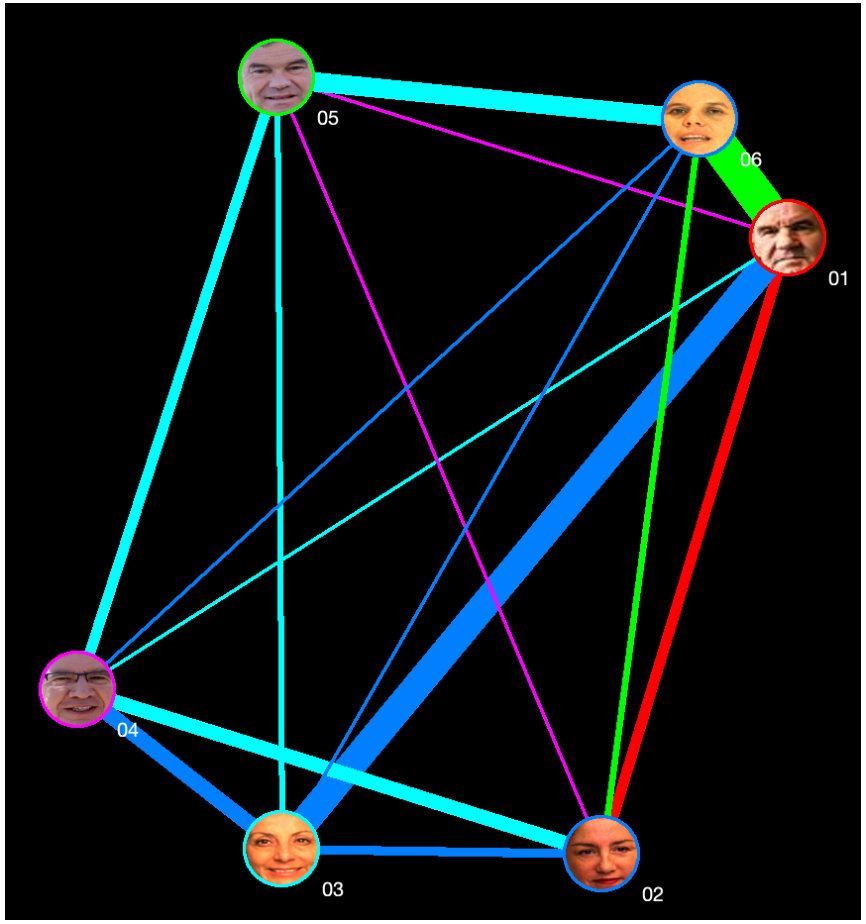
Ideas for more representations

In this representations we could choose

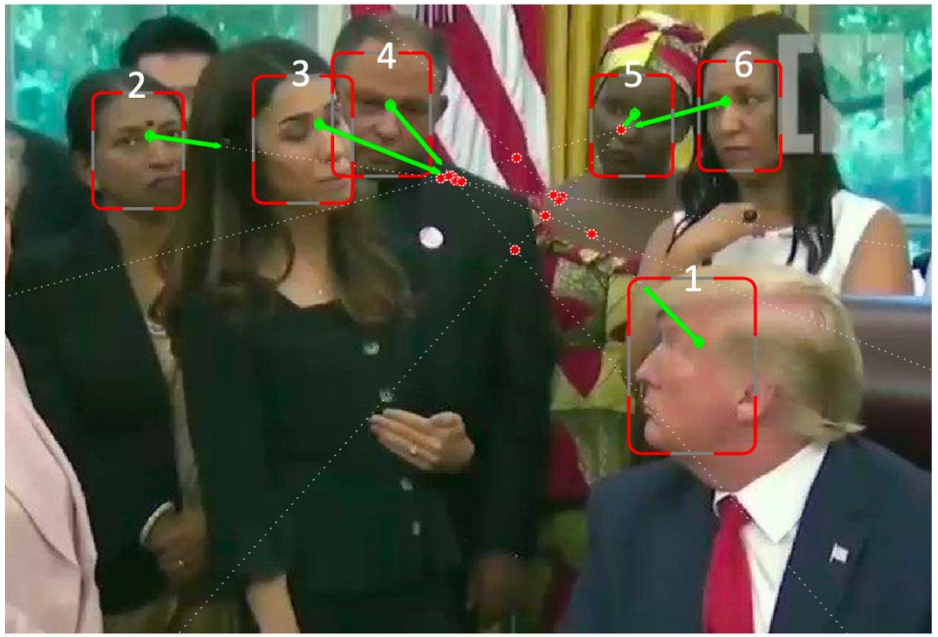
- Length of connections
- Color of connections
- Width of connections
- Size of each face
- Boundary color of each face

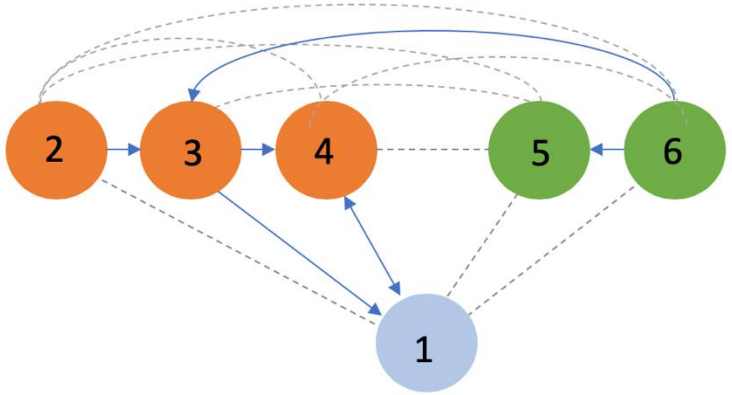
For example:

- Color of connection could be related to facial expression
- Width of connection related to co-occurrence
- Length of connection related to distance in pixels



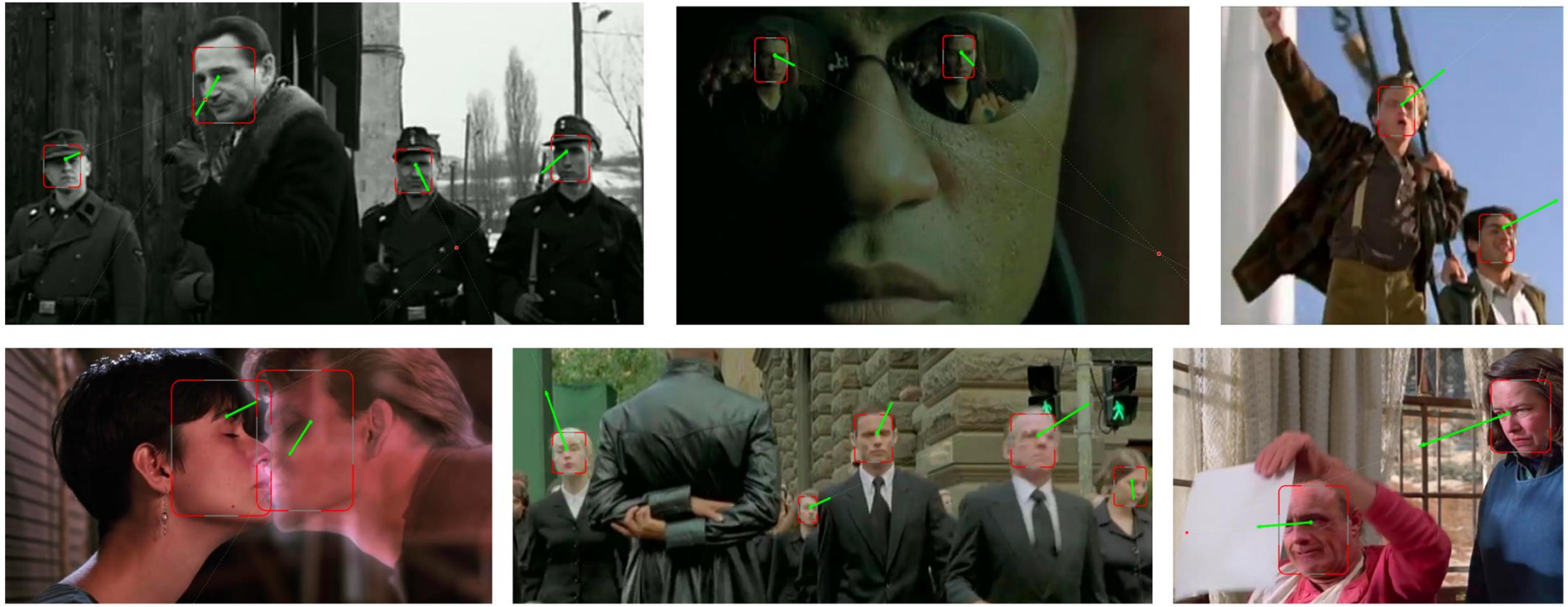
Interactions



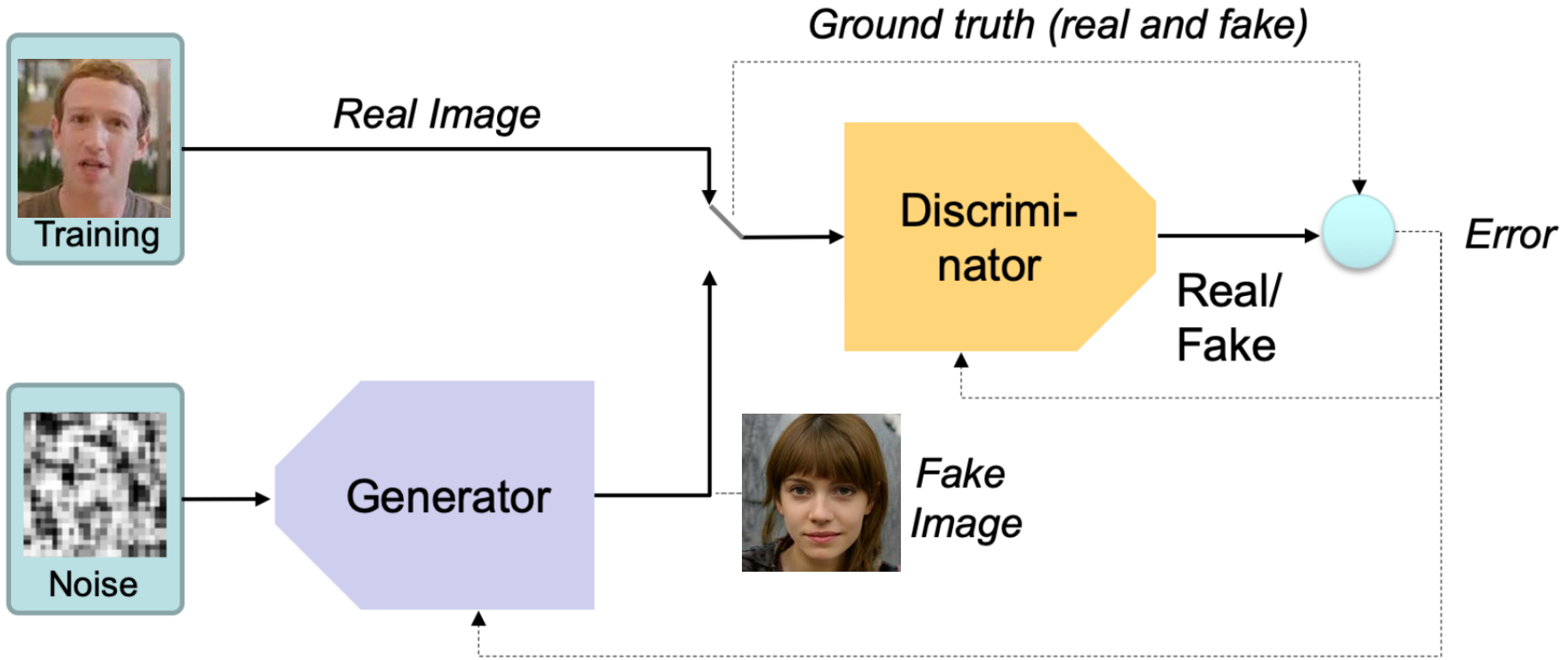


Person	Angry	Disgust	Scared	Happy	Sad	Surprised	Neutral
1	8%	0%	19%	10%	19%	7%	36%
2	16%	1%	21%	8%	16%	2%	37%
3	9%	1%	14%	42%	10%	4%	20%
4	5%	0%	6%	55%	5%	2%	27%
5	11%	1%	13%	34%	6%	2%	34%
6	7%	0%	6%	14%	14%	3%	54%

Interactions



10. GAN / Deep Fakes (BONUS)



www.ThisPearsonDoesNotExist.com





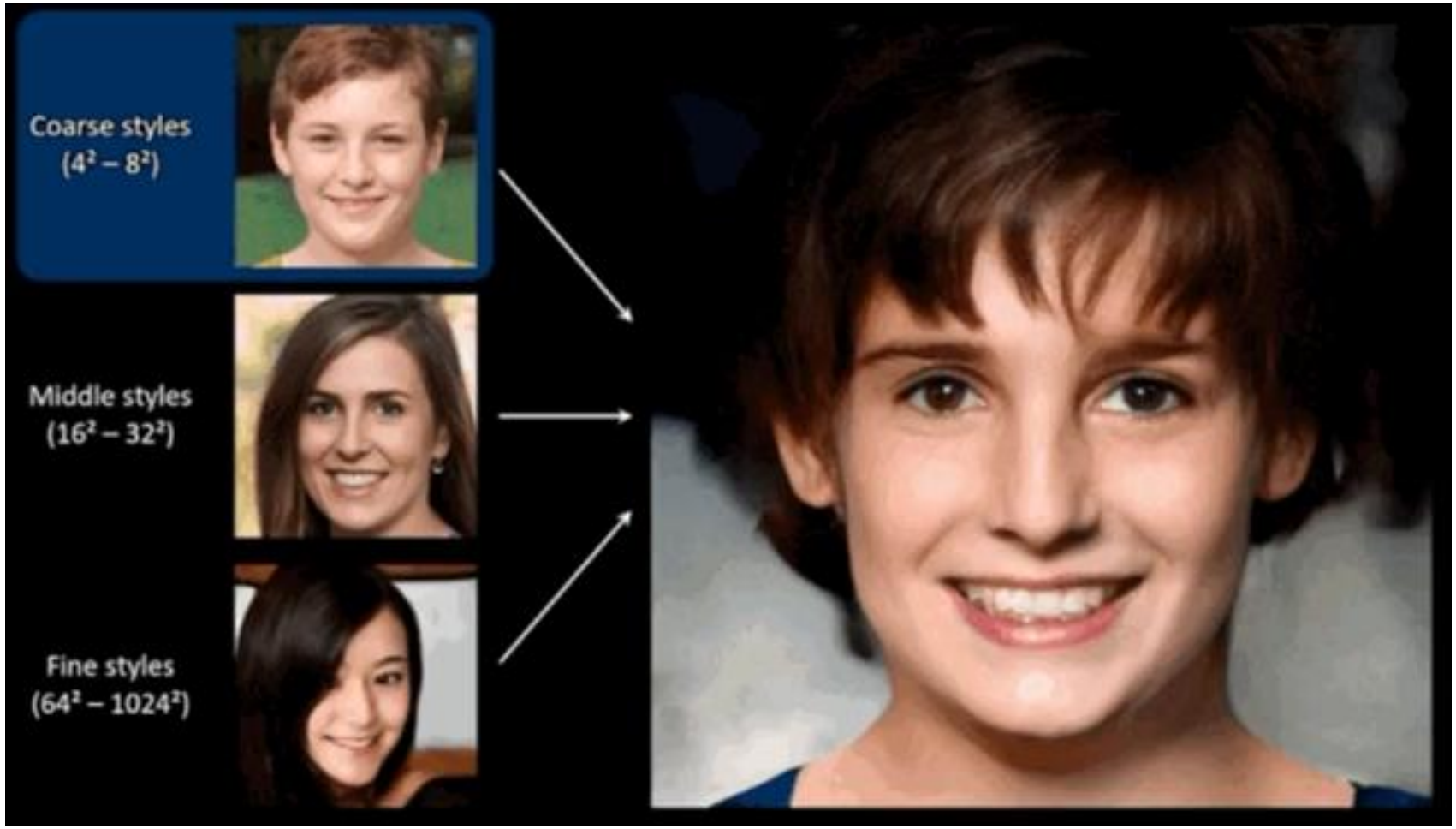




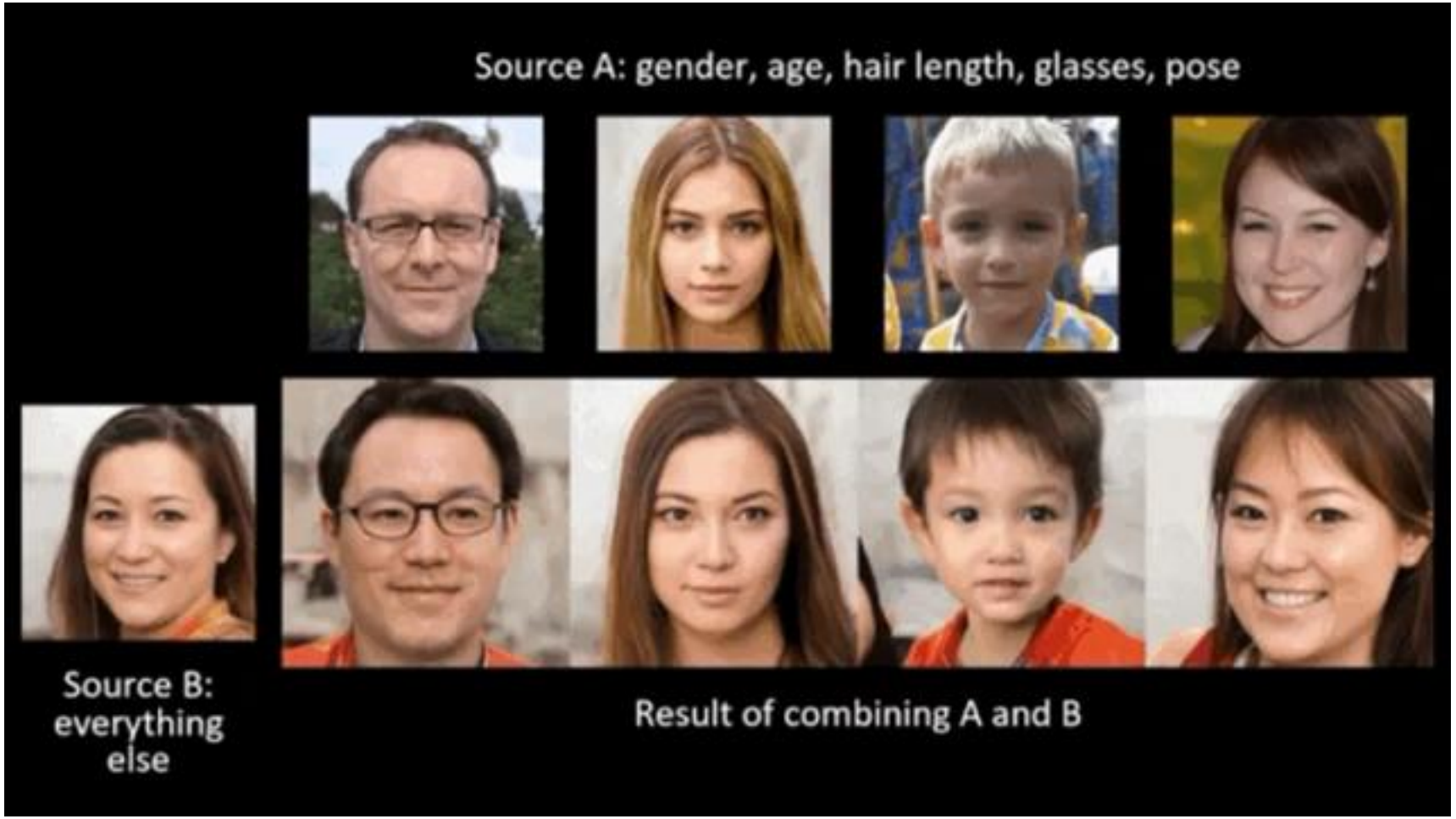








https://syncedreview.com/2019/02/09/nvidia-open-sources-hyper-realistic-face-generator-stylegan/



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8. (

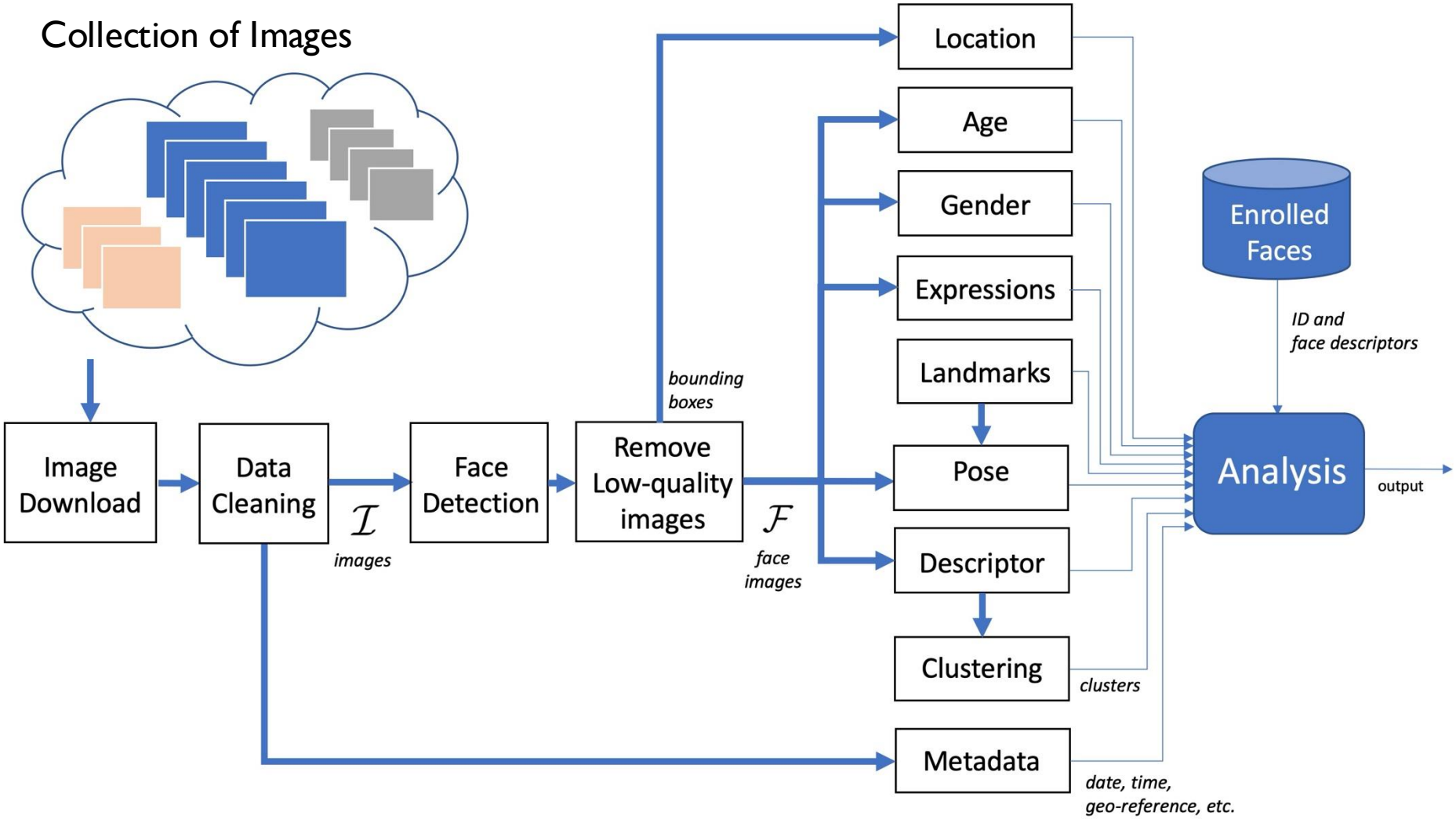


FakeApp: ¿cómo hacer un DeepFake y cómo funciona la app? - Cultura Geek

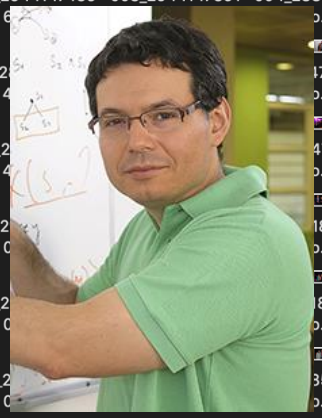


8 steps to verify Deep Fake videos | by Konrad Weber | Medium

Análisis Facial



Buscando a Jorge



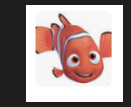
Database with 1122 pictures

Buscando a Jorge

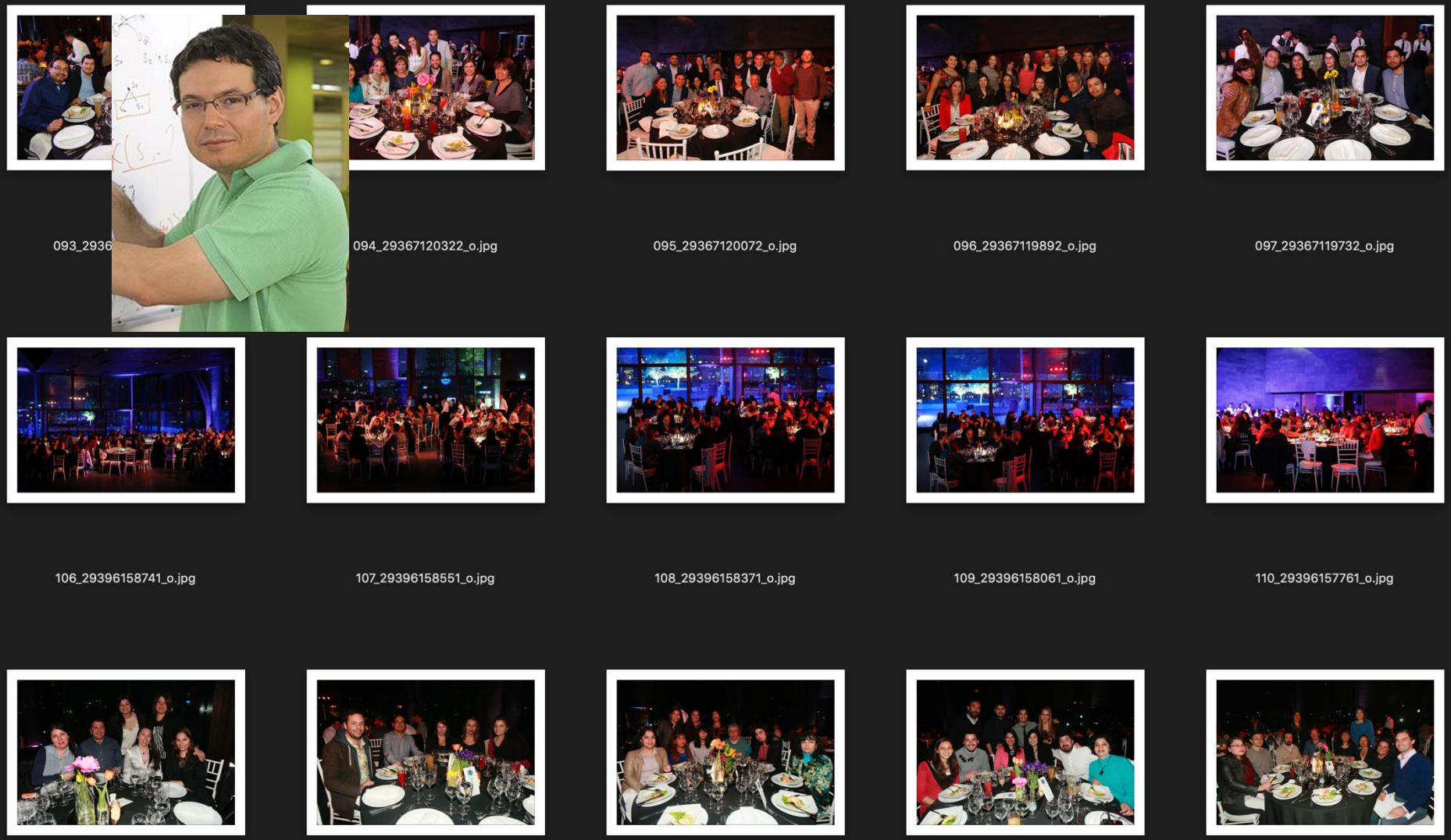
Database with 1122 pictures



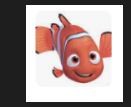
Buscando a Jorge



Database with 1122 pictures



Buscando a Jorge



Database with 1122 pictures

